The Unintended Consequences of Carbon Markets

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Abstract

This paper shows that carbon markets inadvertently weaken incentives for green innovation in hard-to-abate industries by allowing firms to rely on purchasing emission allowances rather than independently decarbonizing. A triple-difference analysis of the European Union Emissions Trading System (EU-ETS) finds that the rise in green patenting is concentrated among easy-to-abate firms, with no comparable increase among hard-to-abate firms. A directed technical change model of a carbon market with sectoral differences in abatement costs shows that carbon markets inefficiently direct most innovation efforts toward decarbonizing the easy-to-abate sector. Socially optimal sector-specific research subsidies redirect innovation to support a clean transition in the hard-to-abate sector and increase welfare by 3.8% in consumption-equivalent terms. These subsidies need to be nearly three times as large (in present value) as those provided to innovators in the easy-to-abate sector.

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1 Introduction

Carbon pricing has been established as a key mechanism in the global decarbonization effort. To curb climate change, governments have implemented a variety of carbon pricing mechanisms, such as carbon taxes and carbon markets. The latter have emerged as the predominant approach, covering over 20% of global emissions, which are more than three-quarters of all priced emissions (World Bank, 2025). Carbon markets operate by enforcing an emissions cap and allowing firms to trade emission allowances. Emissions trading relies on differences in abatement costs between firms and industries. Firms in hard-to-abate industries (such as steel, cement, and chemicals) can purchase allowances from firms in easier-to-abate industries (such as power generation and light industry), creating gains from trade and concentrating emissions reduction where it is most cost-effective.

However, the very structure that facilitates cost-efficiency may also create a dynamic inefficiency. Innovation is subject to a well-studied market failure arising from knowledge spillovers. Since knowledge is non-rival and only partially excludable, innovators cannot entirely prevent others from using and building on their inventions, and thus cannot fully capture their economic value. This wedge between the social and private return to innovation is widened by carbon trading. Because participants can comply by purchasing emission allowances, hard-to-abate firms may meet regulatory obligations without decarbonizing their own operations, weakening their incentives to develop clean production methods. The resulting misallocation of innovation could undermine the efficiency of the clean transition and jeopardize timely progress toward climate neutrality. Despite the widespread adoption of carbon markets, this distortion remains unexplored.

This paper fills this gap by investigating, empirically and theoretically, the differential effect of carbon markets on green innovation and decarbonization across easy- and hard-to-abate industries. The empirical analysis focuses on the European Union Emissions Trading System (EU-ETS), which is the first and largest carbon market. I use a triple-difference approach to analyze how carbon markets affect innovation in easy- versus hard-to-abate industries. Patenting activity serves as a measure of innovation, with "green" patents identified using the CPC's Y02 climate-mitigation category and restricted to technologies relevant to the filing firm's own abatement category (easy- or hard-to-abate), isolating the innovation the carbon market directly incentivizes: a firm's development of clean technology to decarbonize its own production. I compare green innovation activity of firms regulated by the EU-ETS (EU-ETS firms) to those that are not (non EU-

ETS firms) across easy- and hard-to-abate industries, in the pre- and post-policy periods. The findings show that the effect of the EU-ETS on innovation is centered in easy-to-abate firms, with no comparable increase among hard-to-abate firms. Using forward citations as a measure of innovation quality, I document a similar heterogeneous effect on the quality of green innovation. These findings are robust to alternative definitions of green patenting, the inclusion of country-industry-year fixed effects, a continuous measure of abatement cost, and additional tests (Section 4.1). Finally, installations operated by easy-to-abate firms reduced emissions at an annual rate more than four times as high as that of their hard-to-abate counterparts. Taken together, the empirical results show that carbon markets concentrate both innovation and decarbonization in easy-to-abate industries.

The reduced-form analysis alone cannot speak to the welfare implications of the results. I therefore develop a directed technical change model to assess how the structure of carbon markets influences the efficiency of the global decarbonization effort in the presence of sectoral heterogeneity in abatement costs. The model builds on the canonical framework of Acemoglu et al. (2012). A representative household consumes a final good composed of two complementary intermediate goods: an easy-to-abate and a hard-to-abate good, which are produced in separate sectors that differ in their abatement costs. Each sector uses a mix of two substitutable inputs: a common 'dirty' input (fossil fuels) and a sector-specific 'clean' input (non-emitting production methods). The use of the dirty input generates carbon emissions, exacerbating global temperature increase and reducing final output. Each input is produced using labor and machines, with each machine embodying a technology that determines its productivity. I model the difference in abatement costs as arising from an initial gap in the sectors' clean technologies. Scientists, representing innovation efforts, consider expected profits and decide each period in which of the three input sectors to innovate (the dirty input, or one of the two sector-specific clean ones). Successful scientists improve the productivity of their assigned machine. Through a *building on the shoulders of giants* feature, this improvement spills over, raising the technological base on which future innovation builds and creating a positive knowledge externality the scientist does not internalize. Climate policy operates through a declining emissions cap enforced by a carbon market. The rate at which the cap decreases reflects the stringency of the policy and determines the pace of decarbonization. Each period, producers of intermediate goods are subject to an emissions cap and can trade emission allowances, generating a market price for carbon.

I then calibrate the model to analyze the dynamics of the technological transition un-

der (1) a carbon market whose stringency matches climate neutrality by 2050 ambitions, with (2) the transition under the same conditions supplemented by optimally designed sector-specific research subsidies, which are set to maximize welfare given the carbon market's stringency. The results of the first scenario provide theoretical support for the empirical findings: carbon markets disproportionately direct innovation toward the clean easy-to-abate sector, leaving clean hard-to-abate production technology underdeveloped. Furthermore, the calibration predicts a slower transition in the hard-to-abate sector. In the second scenario, research subsidies can be used to increase expected returns for innovators in a given sector. The addition of these subsidies shows that the dynamics of the first scenario are inefficient. The policymaker uses research subsidies to immediately draw innovation efforts to the clean hard-to-abate sector, mitigating the dynamic inefficiency that arises as carbon trading widens the gap between the social and private return to clean innovation in that sector. These subsidies need to be 170% higher in present value than those provided to scientists innovating in the clean easy-to-abate sector (up to 239% across specifications). By advancing hard-to-abate clean technology, the subsidies reduce the sector's allowance purchases by 75% and accelerate its decarbonization and transition to clean production. As a result, a prolonged stagnation in hard-to-abate intermediate good production, driven by stringent climate policy and a lag in its clean technology, is mitigated, yielding welfare gains of 3.8% in consumption-equivalent terms (up to 5.8% across specifications).

These findings show that carbon markets inadvertently provide inefficiently weak incentives for green innovation in hard-to-abate industries. Although they achieve cost-effective emissions reduction, carbon markets alone cannot ensure an efficient transition to clean production, as they exacerbate the market failure of innovation in the hard-to-abate sector. Policymakers should therefore complement them with support for clean innovation in these industries. Yet climate finance flows may point in the opposite direction: as shown in Table 1, in recent years (2019-2022) easy-to-abate industries ("Energy Systems" in the table) received much more public investment, with \$12.8 invested per ton of CO_2 emissions compared to just \$0.9 per ton for hard-to-abate industries ("Industry"). Correcting this disparity is critical to fostering an efficient path to climate neutrality.

This paper contributes to several strands of literature. First, it relates to the literature on directed technical change and the environment. Acemoglu (2002) studies how economic incentives shape the direction of innovation, building on the endogenous growth literature (Romer, 1990; Aghion and Howitt, 1992). Acemoglu et al. (2012) and Acemoglu

et al. (2016) apply this framework to study innovation and sustainable growth in the presence of environmental externalities. They find that short-lived research subsidies, paired with carbon taxes, can trigger a transition to clean production, thereby preventing an environmental disaster.[1] Aghion et al. (2025) study green technological transition along the supply chain, showing that even with Pigouvian taxes, targeted industrial policies are essential to avoiding inefficient steady states. Jo and Karydas (2022) show that climate policy induces the exit of emission-intensive firms, increasing the average elasticity of substitution between clean and dirty inputs and lowering the carbon tax required for decarbonization. Another stream characterizes different mechanisms behind directed innovation, showing that relative prices, energy scarcity, and the elasticity of substitution between clean and dirty inputs govern whether inventive effort can be shifted toward cleaner or more energy-saving technologies (André and Smulders, 2014; Shanker and Stern, 2018; Hassler et al., 2021; Casey, 2024; Alfaro et al., 2025). This paper contributes to this literature by examining how carbon markets, the predominant carbon pricing mechanism, affect the direction of clean innovation. I develop a directed technical change model with a cap-and-trade system and show that emission allowance trading, combined with sectoral differences in abatement costs, inefficiently channels innovation resources into the easy-to-abate sector. The paper underscores the importance of setting *sector-specific* research subsidies, as targeted clean research subsidies to hard-to-abate industries can offset this distortion.

Second, the paper relates to the literature on environmental policy and the direction of innovation. Clean technology is subject to two market failures at once: the full social cost of emissions is not borne by emitters, and the full social return to innovation is not captured by inventors due to knowledge spillovers, leaving it underprovided in the laissez-faire equilibrium (Jaffe et al., 2005). A large body of work finds that environmental regulation and energy prices indeed direct innovation toward cleaner and more energy-saving technologies across a wide range of contexts (Jaffe and Palmer, 1997; Popp, 2002; Newell et al., 2006; Popp, 2006; Noailly and Smeets, 2015; Aghion et al., 2016; Brown et al., 2022. See Popp et al. (2010) and Popp (2019) for reviews of this literature.). This paper is related mainly to studies of the effect of emissions trading schemes on green innovation. Calel and Dechezleprêtre (2016) and Calel (2020) provide causal evidence that the EU-ETS increased low-carbon innovation among regulated firms, and Cui et al. (2018), Zhu

[1] Additional work in this literature includes Fischer and Newell (2008), Acemoglu et al. (2014), Hémous (2016), Van den Bijgaart (2017), Fried (2018), Acemoglu et al. (2023), Lemoine (2024). For reviews of the literature, see Hémous and Olsen (2021) and Dechezleprêtre and Hémous (2022).

et al. (2019) and Weng et al. (2022) find similar results in the context of the Chinese carbon market.[2] This paper extends this literature by showing that while the EU-ETS leads to an overall increase in green innovation, the effect is inefficiently weak in hard-to-abate industries.

Third, the paper contributes to the literature on the effectiveness of emissions trading schemes. Montgomery (1972) provides the theoretical foundations for the efficiency of emissions trading, showing that, for a given cap, a competitive market in tradable allowances meets the emissions target at minimum aggregate cost by equalizing marginal abatement costs across firms, independently of the initial allocation of allowances. Estimating marginal abatement costs in the US sulfur dioxide market, Carlson et al. (2000) quantify the long-run gains from allowance trading at $700–800 million per year relative to a command-and-control alternative prescribing uniform emission standards. Weitzman (1974) shows that under uncertainty about abatement costs, a quantity instrument such as a cap-and-trade market is preferred to a price instrument when the marginal benefit of abatement is steep relative to its marginal cost. Fabra and Reguant (2014) show that electricity producers fully internalize the opportunity cost of freely allocated allowances, while Fowlie et al. (2016) show that market power and emissions leakage can render market-based regulation welfare-reducing. The EU-ETS substantially reduced emissions (Bayer and Aklin, 2020), with regulated firms abating without harm to their economic performance (Dechezleprêtre et al., 2023; Colmer et al., 2024), yet firms' compliance behavior can undermine these gains, whether by selling surplus allowances after voluntary reductions or by relying on low-quality offsets (Akey et al., 2024; Bellon et al., 2026). I document that EU-ETS emissions reductions are concentrated in easy-to-abate industries, achieving static cost-effectiveness, while the same concentration in induced innovation creates a dynamic inefficiency: under-investment in clean technologies for hard-to-abate production.

The remainder of the paper is organized as follows. Section 2 provides institutional background on the EU-ETS and easy- and hard-to-abate industries. Section 3 outlines the data used in the empirical analysis, and Section 4 presents the empirical framework and results. Section 5 introduces the theoretical model, Section 6 presents calibration results, and Section 7 concludes.

[2]See also Lucena-Giraldo et al. (2022) and Ren et al. (2022), and Rastegar et al. (2024) for a literature review.

2 Institutional Background

2.1 The EU Emissions Trading System

The European Union Emissions Trading System (EU-ETS) is the first and largest carbon market currently operating. Following a pilot phase (2005-2007), the market began full operation in 2008 as the cornerstone of the EU's climate policy. It covers emissions from power plants, industrial facilities, aviation (since 2012), and maritime transport (since 2024), which together account for approximately 40% of the EU's total greenhouse gas emissions. Installations from all EU countries participate, as well as installations from the United Kingdom (until 2020), Iceland, Liechtenstein, and Norway. Under this cap-and-trade system, participants receive or purchase allowances to emit CO_2, and those that emit less than their allowances can sell the surplus to firms facing higher abatement costs, promoting a cost-effective allocation of emissions reductions. Firms exceeding their allowances must purchase additional ones or face penalties that have always been higher than the market price of allowances (see European Commission (2024), Chapter 2, for more details).

The introduction of the EU Emissions Trading System (EU-ETS) marked a transformative shift in EU climate policy, replacing the fragmented and incomplete landscape of sector-specific and country-specific instruments with a unified mechanism. Before the EU-ETS, Denmark, Finland, the Netherlands, Norway, Poland, and Sweden implemented carbon taxes in the 1990s, targeting various industries (Martinsson et al., 2024). In contrast, many member states, such as France, Spain and the UK, did not implement any carbon pricing mechanism prior to the establishment of the EU-ETS (see Online Appendix A of Metcalf and Stock (2023)). The European Commission's debates on the EU's Kyoto Protocol commitment to reduce GHG emissions by 8% by 2012 (compared to 1990 levels) highlight the limitations of disparate national policies. A harmonized, EU-wide emissions trading system was viewed as a superior approach to ensuring cost-effective and coordinated compliance.[3]

Inclusion in the EU-ETS is based on production capacity thresholds, capturing large emitters while balancing administrative efficiency. For example, power plants and industrial installations with a thermal input greater than 20 megawatts are included, along with iron and steel producers exceeding production capacity of 2.5 tons of pig iron or steel per

[3]See Document 52000DC0087 dated March 8, 2000, on EUR-Lex (Accessed on 01/20/2025).

hour.[4]

Initially, most allowances were freely allocated to ease the transition and mitigate carbon leakage, the risk of production moving to regions with weaker climate policies. Over time, the system shifted toward auction-based allocation. However, to maintain global competitiveness, participants in hard-to-abate industries continue to receive a diminishing share of free allowances, which remain substantial and dampen the policy's effective stringency.[5] This factor is a potential confounder in my analysis, which I address by controlling for the emissions-to-free-allocation ratio, as discussed in Section 4.

Now in its fourth phase (2021-2030), the EU-ETS continues to evolve, with reforms aimed at reducing free allowances and aligning the system with the European Green Deal. The inclusion of additional sectors, such as maritime transport (starting in January 2024), reflects ongoing efforts to expand the scope of the EU-ETS and support the EU's goal of achieving climate neutrality by 2050.

The design of the EU-ETS guides the triple-difference analysis in this paper. First, the structure of the inclusion criteria creates within-industry variation between regulated and unregulated firms, providing the basis for a difference-in-differences analysis (Calel and Dechezleprêtre, 2016; Dechezleprêtre et al., 2023), comparing EU-ETS and non EU-ETS firms, before and after the introduction of the market. Together with the inclusion of a wide range of industries with differences in the cost of abatement, the EU-ETS is well-suited for a triple-difference analysis that studies the heterogeneous effect of the EU-ETS on easy- versus hard-to-abate industries.

2.2 Easy- and Hard-to-Abate Industries

Industries can be broadly categorized based on their abatement potential, reflecting the ease or difficulty of reducing greenhouse gas emissions in their operations. Hard-to-abate sectors, such as steel, cement, chemicals, oil, and heavy transportation, account for roughly 40% of global emissions (World Economic Forum, 2024). Their reliance on energy-intensive processes and fossil fuels makes decarbonization particularly challenging. In contrast, easy-to-abate sectors, such as electricity generation and light industry,

[4]The complete list of inclusion criteria and activity-level thresholds (when applicable) can be found in Directive 2003/87/EC of the European Parliament and of the Council and its amendments (Accessed on 01/21/2025).

[5]See Commission Decision 2011/278/EU (Accessed on 01/20/2025) dated April 27, 2011, for further details.

emit less per unit of output and can reduce emissions cost-effectively with existing technologies.

The fundamental difference between easy- and hard-to-abate industries lies in the nature of their emissions and the feasibility of reducing them. Hard-to-abate producers, and particularly heavy manufacturers, often emit *process* emissions, which are byproducts of chemical reactions in production, such as the release of CO_2 from calcination in cement production or from iron ore reduction (into pig iron) in steelmaking. These emissions cannot simply be reduced by switching to renewable energy or improving efficiency, and require the development of breakthrough technologies such as non-lime based cement production methods or green hydrogen-based steelmaking (Appendix A provides a detailed discussion of the technological landscape of the largest hard-to-abate industries). By contrast, easy-to-abate industries can reduce emissions by switching to renewables, which experienced rapid development in the previous decades. In this paper I show that this distinction is critical for climate policy, as carbon markets tend to incentivize accelerated emissions reduction and increased green innovation in easy-to-abate sectors, leading to weaker incentives for decarbonization and innovation in hard-to-abate industries.

3 Data

This section details the data sources and the construction of the sample. The main sample used in the empirical section includes information on innovation activity of European firms in 2000-2020 in key industries that are regulated by the EU-ETS.

3.1 Data Sources

I construct the sample using two main data sources: (1) the EU Transaction Log (EUTL) provides information on participation in the EU-ETS and installation-level emissions; and (2) the World Patent Statistical Database (PATSTAT) provides patent information that is used to construct firm-level measures of green innovation activity.

EU Transaction Log (EUTL) The European Union Transaction Log (EUTL) tracks compliance behavior of installations that participate in the EU-ETS.[6] For each installation, the

[6]Data are accessed through euets.info. See Abrell (2021) and https://union-registry-data.ec.europa.eu/report/welcome for further details.

EUTL documents the yearly emissions and the amount of emission allowances that are freely allocated. In addition, the EUTL includes information on the accounts that manage the compliance of installations, allowing for a match with other data sources at the firm level. This provides a comprehensive list of all firms that operate EU-ETS installations, as well as information on annual emissions and free allocation at the installation level.

Patent Data I use patenting as a measure of innovation activity, with the European Patent Office (EPO)'s PATSTAT serving as the primary data source (European Patent Office, 2025). I extract all patents filed with the EPO by firms registered in countries participating in the EU-ETS between 2000 and 2020. I use EPO patent filings because they provide a harmonized cross-country measure of innovation, and are not prone to potential standard differences between national patent offices. They also serve as an ex ante quality screen, since higher-value inventions are more likely to be filed at the EPO, helping screen out lower-quality patents (Calel and Dechezleprêtre, 2016; Aghion et al., 2016). To identify "green" patents, I follow established literature (e.g., Calel and Dechezleprêtre, 2016) and use the Y02 category of the CPC classification, which covers "technologies or applications for mitigation or adaptation against climate change" (Cooperative Patent Classification, 2025). I focus on Y02 subcategories related to electricity generation, energy efficiency, production processes, and carbon capture, utilization, and storage (CCUS).[7]

The first-order effect of the carbon market is to induce innovation in clean production processes directly relevant to a firm's core economic activities, since it covers direct (Scope 1) emissions. I therefore adopt a technology-specific definition of green patenting as my baseline: a patent filed by an easy-to-abate (hard-to-abate) firm is classified as "green" only if it pertains to a technological solution relevant to an easy-to-abate (hard-to-abate) economic activity. I classify the CPC Y02 sub-categories into those that address easy- and hard-to-abate economic activities.[8] For example, technologies under "energy

[7] I classify patents as green if they fall under sub-categories Y02B (climate change mitigation technologies related to buildings), Y02C (capture, storage, sequestration, or disposal of greenhouse gases), Y02D (climate change mitigation technologies in information and communication technologies), Y02E (reduction of greenhouse gas emissions related to energy generation, transmission, or distribution), or Y02P (climate change mitigation technologies in the production or processing of goods). I exclude the remaining subcategories, Y02A (adaptation to climate change), Y02T (transportation), and Y02W (wastewater treatment or waste management), since they do not relate to abatement of direct emissions of the patenting firm. For example, Y02T patents cover sustainable transportation such as electric vehicles, which reduce emissions in use rather than in the patenting firm's own production.

[8] I classify each of the Y02B, Y02C, Y02D, Y02E, and Y02P CPC subcategories as predominantly relevant to hard-to-abate economic activities (steel, cement, chemicals, additional heavy manufacturing, and oil refining and extraction) or to easy-to-abate economic activities (light industry and electricity generation),

generation through renewable energy sources" (Y02E 10/00) are predominantly relevant to easy-to-abate activities because they help reduce direct emissions from electricity production. In contrast, patents classified under "technologies related to the processing of minerals" (Y02P 30/00) provide emissions-reducing solutions for cement, concrete, or glass manufacturing, which are hard-to-abate economic activities. Table A.1 presents the full classification used in the baseline analysis. In Section 4.1, I show that the results are robust to broader definitions that relax the own-technology requirement.

3.2 Matching and Sample Construction

Matching To create a firm-level panel dataset, including patenting activity and classification of firms by exposure to climate policy through the EU-ETS, I match the patent data with the EUTL data. I first transform the patent data from patent level into firm-year level using the OECD Harmonized Applicant Name database (HAN database) that is constructed to standardize names of patentees for the purpose of matching PATSTAT to firm-level data (OECD, 2020). To avoid double counting of an invention that is filed multiple times (in different countries or due to slight variations of the technology), patent applications are grouped into patent families (Dugoua and Gerarden, 2025). The patent data are then matched to a list of Orbis Large or Very Large firms ((1) operational revenue $\geq$ €10 million; (2) total assets $\geq$ €20 million; or (3) number of employees $\geq$ 150), to avoid over-representation of small firms in the non EU-ETS firm sub-sample due to the inclusion criteria of EU-ETS discussed in Section 2.1 (Bureau van Dijk, 2024). The matching process involves pre-cleaning and standardization of firm names, followed by a stepwise matching procedure using exact and fuzzy matching techniques, employing LinkTransformer (Arora and Dell, 2023), a large language model (LLM) record linkage tool. Next, the firm-level panel is matched to the EUTL data, primarily relying on the crosswalk between EUTL and Orbis constructed by the Joint Research Centre (JRC) of the European Commission (Letout et al., 2021). This is supplemented with the matching algorithm used to link PATSTAT and Orbis.

Sample Construction The main sample is constructed in the following steps. First, firms are considered to be in the treatment group if they operate at least one installa-

based on a detailed study of the underlying technologies and the industries in which they are applied. I then independently verify each assignment using a large language model (OpenAI's ChatGPT), and treat any subcategory for which the two classifications disagree as unclassified.

tion that participates in the EU-ETS. This definition follows the literature (e.g., Calel and Dechezleprêtre, 2016) and enables me to separate firms that are directly affected by the introduction of the EU-ETS from those that are not. To standardize firms across the treatment and control groups in terms of their patenting activity, I exclude firms that did not file at least one EPO patent in the pre-period (2000-2007). A robustness check varies this restriction, both relaxing and tightening it, and shows that the results hold (Appendix B). Second, I exclude non EU-ETS firms that are parents, sisters or subsidiaries of EU-ETS firms (Calel and Dechezleprêtre, 2016). Third, I exclude industries (at the 3-digit NACE code level) for which there are no firms in the control or treatment groups. Finally, following the discussion in Section 2.2, I consider power and light industry companies as easy-to-abate, and heavy industry and the oil industry as hard-to-abate.[9] The results are robust to replacing this binary classification with a continuous measure of industry emissions intensity (Section 4.1). Throughout, the sample is restricted to Orbis Large and Very Large firms (operational revenue of at least €10 million, total assets of at least €20 million, or at least 150 employees; Bureau van Dijk (2024)). Because inclusion in the EU-ETS is determined by the production-capacity thresholds discussed in Section 2.1, this restriction alleviates the concern that treated firms would otherwise be systematically larger than the firms in the control group. The final sample includes 1,969 firms in 2000-2020, of which 389 firms are in the treatment group (EU-ETS firms). Tables 2 and 3 present the number of firms by country and industry, respectively.

4 Empirical Analysis

I start by documenting two stylized facts. First, easy-to-abate EU-ETS installations reduce emissions at more than four times the annual rate of their hard-to-abate counterparts. Second, the post-policy increase in green innovation is concentrated among easy-to-abate firms. These facts point to the two-sided effect of carbon markets. On the one hand, they promote cost-effective emissions reduction through carbon trading; on the other, they

[9]Specifically, I consider NACE categories D35 (Electricity, gas, steam and air conditioning supply), C10 (Manufacture of food products), C11 (Manufacture of beverages), C12 (Manufacture of tobacco products), C29 (Manufacture of motor vehicles, trailers and semi-trailers) and C30 (Manufacture of other transport equipment) as easy-to-abate industries, and NACE categories B (Mining and quarrying), C17 (Manufacture of paper and paper products), C19 (Manufacture of coke and refined petroleum products), C20 (Manufacture of chemicals and chemical products), C23 (Manufacture of other non-metallic mineral products), C24 (Manufacture of basic metals) as hard-to-abate industries. Other industries are excluded from this study due to low participation in the EU-ETS.

create weaker incentives for hard-to-abate firms to pursue the technological innovations necessary for clean production.

To analyze emissions reduction, I construct a panel dataset of all installations associated with the firms in the main sample constructed in Section 3.2, and estimate an event-study specification separately for easy- and hard-to-abate installations:

$$\log(emissions_{it}) = \alpha_i + \sum_{\tau=1}^{10} \beta_\tau \cdot I\{\text{event time}_{it} = \tau\} + \varepsilon_{it}, \tag{1}$$

where α_i denotes installation fixed effects and event time measures the number of years since installation i's inclusion in the EU-ETS, with the inclusion year ($\tau = 0$) as the omitted baseline. Standard errors are clustered at the installation level. The coefficients β_τ trace the change in emissions relative to the inclusion year.

Figure 1 presents the results. Emissions of easy-to-abate installations decline steadily, falling by about 30% within ten years of inclusion, whereas emissions of hard-to-abate installations fall by less than 10% over the same horizon, and the two trajectories diverge from about the third year after inclusion. Appendix B confirms that this divergence is statistically significant. Expressed as average annual rates, these cumulative declines correspond to about 3.5% per year for easy-to-abate installations versus about 0.8% per year for hard-to-abate installations.

The second stylized fact regards green innovation. Figure 2 presents the trend in average *green* patenting per firm, comparing firms regulated by the EU-ETS to those that are not. The figure separates the sample into easy-to-abate industries (Panel A) and hard-to-abate industries (Panel B). Panel A shows a substantial increase in green patenting activity among easy-to-abate firms covered by the EU-ETS following the system's full implementation in 2008, with green patenting rising to roughly four times its 2007 level by 2020. In contrast, hard-to-abate EU-ETS firms show no comparable increase, remaining roughly flat over the same period. This pattern suggests that the EU-ETS may provide much weaker incentives for clean innovation in hard-to-abate industries, motivating the empirical analysis presented below.

To identify the causal heterogeneous effect of the EU-ETS on green innovation, I employ a triple-difference specification:

$$green_{it} = \alpha + \beta \cdot ETS_i \cdot Post_t \cdot Hard_i + \gamma \cdot ETS_i \cdot Post_t + X'_{it}\delta + \phi_i + \theta_{ct} + \lambda_{jt} + \varepsilon_{it} \tag{2}$$

where $green_{it}$ is a measure of *green* innovation conducted by firm i in year t. The main outcome variable is an indicator variable equal to 1 if the firm files at least one green patent in a given year, capturing the extensive margin of green innovation (*engagement*). In Section 4.1, I show that the results are similar when the outcome captures the intensive margin of green innovation (*intensity*), measured as the number of green patents filed. ETS_i is set to 1 for firms that are regulated by the EU-ETS (that is, operate at least one installation that participates in the EU-ETS), $Post_t$ is set to 1 for years after the full operation of the carbon market (2008),[10] and $Hard_i$ is set to 1 for firms that operate in a hard-to-abate industry. The hypothesis is that the coefficient on the triple interaction term is negative $\beta < 0$, indicating that the post-policy difference in green innovation between treated and control firms is smaller within the hard-to-abate sector compared to within the easy-to-abate sector.

As discussed in Section 2.1, the European Commission has been using free allocation of emission allowances to reduce the abatement costs associated with the policy and alleviate the risk of carbon leakage, mainly for hard-to-abate industries. To account for this confounding factor, which in itself may disproportionately weaken green innovation incentives for hard-to-abate firms, I control (in the vector X_{it}) for the lagged emissions-to-free-allocation ratio at the 2-digit (NACE) industry level.[11] The vector X_{it} also includes the lagged log patent stock, which absorbs time-varying differences in firms' innovative capacity.[12]

To isolate the effect of the carbon market on innovation, I use a combination of high-dimensional fixed effects. Firm fixed effects (ϕ_i) are included to account for time-invariant firm characteristics that can influence clean innovation. Country-year fixed effects (θ_{ct}) capture time-varying country-specific determinants of firm green innovation, and specifically control for heterogeneity in energy input prices and supplementary carbon policies at the country-year level (such as the Swedish carbon tax system or other forms of country-level heterogeneity in carbon policy prior to the EU-ETS, as discussed in Section 2.1). Industry-year fixed effects (λ_{jt}) control for industry-specific characteristics or policies that vary over time and may influence green innovation such as the Trans-European

[10]The chemical industry was added in the third phase of the EU-ETS (2013-2020), and therefore this variable is set to 1 after 2013 for firms in the chemical industry (NACE code C20).

[11]For 2000-2007, I assign a value of 0 to this variable, which is conceptually equivalent to having an infinite amount of freely allocated emission allowances, capturing an economy in which carbon emissions are not regulated.

[12]The patent stock is constructed using the perpetual-inventory method with an annual depreciation rate of 0.15 (Hall et al., 2005).

Networks for Energy (TEN-E), which supports the integration of renewable energy in electricity production. The dimensionality of the data does not allow the inclusion of country-industry-year fixed effects in the full sample. In Section 4.1 I present a robustness check with country-industry-year fixed effects restricting to the largest countries (Germany, France, Italy, and the United Kingdom) and show that the findings are unchanged. Standard errors are clustered at the firm level.

Finally, the parallel trends identification assumption in this triple-difference setting is that, in the absence of treatment, the difference in green innovation between treated and control firms would have evolved similarly across the easy- and hard-to-abate sectors (Olden and Møen, 2022). Figure 3 plots the triple interaction coefficients in a yearly event-study specification of Equation 2 (see the figure note for full details), and shows no evidence of pre-trends, supporting the validity of the parallel trends assumption.

Table 4 presents estimation results of Equation 2. In Section 4.1, additional tests and robustness checks are presented. The estimate of the coefficient of interest, β, is negative in all specifications, and particularly when firm, country-year, and industry-year fixed effects are included (column 4, where $\hat{\beta} = -0.063$), indicating that the carbon market provides significantly weaker incentives to engage in green innovation for firms in hard- versus easy-to-abate industries. To illustrate the magnitude of this effect, Table 5 presents estimation results of a difference-in-differences regression splitting the sample by the "third" difference into firms in easy- and hard-to-abate industries. Columns 8 and 9, which include firm, country-year and industry-year fixed effects, show that easy-to-abate regulated firms are 6.7 percentage points more likely to engage in green innovation after 2008. The corresponding effect on firms in hard-to-abate industries is a slight, statistically insignificant 0.5 percentage points. Scaling this result by the unconditional mean of the dependent variable, the effect for the easy-to-abate subsample is equivalent to a 268% increase in the likelihood of engaging in green innovation, compared to a statistically insignificant 17% increase for the hard-to-abate subsample. These results show that the post-policy rise in green innovation is concentrated in easy-to-abate industries, while the response among hard-to-abate firms is not statistically distinguishable from zero. The triple-difference estimate in Table 4 confirms that this difference between the two sectors is statistically significant, underscoring that carbon markets may inadvertently concentrate innovation in the easy-to-abate sector. Finally, column 7 of Table 5 shows that across both sectors, the EU-ETS led to an increase in green innovation, consistent with previous literature (e.g., Calel and Dechezleprêtre, 2016).

4.1 Additional Tests and Robustness Checks

Intensive Margin: The Effect on the Number of Green Patents The impact of carbon markets on the number of green patents filed by hard-to-abate firms is also notably smaller. Column 2 of Table 6 estimates Equation 2 with the natural logarithm of one plus the count of green patents filed by firm i in year t, $\log(1 + \#\,\text{green patents}_{it})$, as the dependent variable. The triple-interaction estimate of -0.056 is close in magnitude to the extensive-margin estimate of -0.063, indicating that the EU-ETS differential between easy- and hard-to-abate firms operates not only on whether firms engage in green innovation, but also on its intensity: the post-2008 increase in green-patent counts among EU-ETS firms, relative to unregulated firms, is significantly smaller in hard- than in easy-to-abate industries.

Quality-Adjusted Patents I next show that the EU-ETS differentially affected not only the quantity of green innovation, but also its quality. Column 3 of Table 6 presents estimation results of Equation 2 with the dependent variable defined as the natural logarithm of one plus the number of forward citations received by the firm's green patents, $\log(1 + \#\,\text{green citations}_{it})$, a common proxy for innovation quality (Hall et al., 2005; Calel and Dechezleprêtre, 2016; Acemoglu et al., 2023).[13] The triple-interaction estimate of -0.14 indicates that the post-2008 increase in citation-weighted green patenting among EU-ETS firms, relative to unregulated firms in the same industries, is significantly smaller in the hard-to-abate sector than in the easy-to-abate sector. This effect is more than twice the magnitude of the green-patent-count estimate (-0.056, Column 2), showing that the differential effect of the EU-ETS extends to quality as well as quantity, and is in fact larger in quality-adjusted terms.

Robustness to the Definition of Green Innovation The baseline outcome classifies a patent as green only if its technology corresponds to the firm's own abatement category, easy- or hard-to-abate, using a detailed mapping of Y02 subcategories to easy- and hard-to-abate technologies. Columns 4 to 6 of Table 6 show that the result does not hinge on this mapping. Column 4 retains the own-technology restriction but assigns subcategories with a coarser split that follows the CPC subclasses.[14] Column 5 broadens the outcome,

[13] I use a four-year rather than the more common five-year forward-citation window in order to retain patents filed in 2020 in the sample.

[14] The easy-to-abate subclasses are Y02B (emissions reductions in buildings), Y02D (energy efficiency in information and communication technologies), and Y02E (clean energy generation). The hard-to-abate sub-

setting it to one if the firm filed any green patent that is not classified exclusively in the other sector, thereby retaining the unclassified green patents that the baseline leaves out while still excluding innovation specific to the opposite sector. Column 6 drops the own-technology restriction entirely, setting the outcome to 1 if the firm filed any green patent, regardless of the abatement category.[15] The triple-interaction estimate remains negative and significant across all three specifications (-0.040, -0.034, and -0.034) and comparable to the baseline of -0.063 (Column 1). It attenuates modestly as the measure broadens, with significance weakening to the 5%, 10%, and 10% levels across columns 4 to 6, consistent with the own-technology definition isolating the first-order margin more sharply.

Large Economies Sub-sample and Country-Industry-Year Fixed Effects While the main specification includes firm, country-year, and industry-year fixed effects, it does not account for yearly shocks to a specific industry in a given country.[16] Since including country-industry-year fixed effects in the full sample is infeasible, the sample is restricted to firms in Germany, France, Italy, and the United Kingdom (the four largest economies), which together represent 74.6% of the main sample firms (see Table 2). Column 7 of Table 6 shows that the main result, with firm, country-year, and industry-year fixed effects, is robust to this restriction, with a triple-interaction estimate of -0.084. Column 8 replaces the country-year and industry-year fixed effects with country-industry-year fixed effects. The estimate is -0.081, confirming that the findings are not driven by unobserved country-industry trends.

Continuous Measure of Abatement Cost The baseline analysis classifies industries into a binary split between easy-to-abate and hard-to-abate sectors. Using data from the International Energy Agency (IEA) on emissions per value added (International Energy Agency, 2023*b*), I replace this binary classification with a continuous measure based on pre-policy emissions intensity as a proxy for industry-level abatement costs. Because the IEA emissions-per-value-added data are unavailable for the energy sector and for the chemical, mining, and paper industries, firms in these industries are excluded from this test. Specifically, each firm is assigned the average emissions intensity of its industry-country cell over 2000-2007. This measure ranges from 0.13 tons of CO_2 per \$1,000 of

classes are Y02C (capture, storage, and sequestration of greenhouse gases) and Y02P (emissions reductions in the production and processing of goods).

[15] Green patents are those classified under subclasses Y02B, Y02C, Y02D, Y02E, or Y02P.

[16] E.g., the German 2006 Hydrogen R&D program; see https://www.now-gmbh.de/en/funding/funding-programmes/hydrogen-and-fuel-cell/ (accessed 01/24/2025).

value added for transport equipment, the least carbon-intensive industry, to 2.86 for basic metals, the most carbon-intensive industry. The production-weighted average is 0.22 for easy-to-abate industries and 1.35 for hard-to-abate industries, as defined in Section 3.2. Column 9 of Table 6 reports the corresponding triple-interaction estimate of -0.029, indicating that the post-2008 increase in green innovation among EU-ETS firms, relative to unregulated firms, is smaller in more carbon-intensive industries. The heterogeneous effect of the EU-ETS is therefore not an artifact of the binary industry classification.

5 A Directed Technical Change Model of a Carbon Market

The empirical results show that (1) carbon markets provide weaker incentives for green innovation among hard-to-abate firms, and (2) hard-to-abate firms reduce emissions at a slower pace. However, the reduced-form analysis alone cannot speak to the implications of the findings on the efficiency of the transition to clean technology under a carbon market. Therefore, I develop a directed technical change model of an economy governed by a carbon market with heterogeneous abatement costs.

The model allows me to examine the interaction between carbon markets and sector-specific research subsidies, which are widely studied in the literature as a complementary tool to carbon pricing (e.g., Fischer and Newell, 2008; Acemoglu et al., 2012, 2016). The model calibration compares two scenarios: (1) an economy governed by a carbon market whose stringency matches climate neutrality by 2050 ambitions, without research subsidies; and (2) an economy governed by the same carbon market with the addition of welfare-maximizing sector-specific research subsidies. Under the first scenario, clean innovation is centered in the easy-to-abate sector since hard-to-abate producers face weaker incentives for decarbonization, and rely on purchasing emission allowances from their easy-to-abate counterparts. This is in line with the empirical findings. However, optimal research subsidies realign innovation incentives and direct clean innovation toward the hard-to-abate sector, implying that the empirical findings point to an inefficiency of carbon markets.

5.1 Model Setup

The model operates in discrete time and infinite horizon, drawing on Acemoglu et al. (2012). Figure 4 provides a schematic representation.

Representative Household The economy consists of a representative household that derives utility from consuming a unique final good, C_t. Social welfare is given by

$$\sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} \frac{C_t^{1-\sigma}}{1-\sigma}, \tag{3}$$

where $\rho > 0$ is the pure rate of time preference and $\sigma > 0$ is the coefficient of relative risk aversion.

Let S_t denote the atmospheric CO_2 concentration. Following Golosov et al. (2014) and Acemoglu et al. (2023), the stock of atmospheric carbon does not enter utility directly. Instead, it affects welfare indirectly through its effect on final output.

Final Good The unique final good is produced competitively using two intermediate goods: a hard-to-abate good, Y_t^H, representing industries such as steel, cement, and chemicals, and an easy-to-abate good, Y_t^E, representing electricity generation and light industry. Gross output is given by the CES production function

$$\tilde{Y}_t = \left((Y_t^E)^{(\nu-1)/\nu} + (Y_t^H)^{(\nu-1)/\nu} \right)^{\nu/(\nu-1)}, \tag{4}$$

where the intermediate goods are complements, $\nu < 1$.

Adopting the damage specification of Golosov et al. (2014), climate damages reduce final output according to

$$Y_t = (1 - D(S_t))\, \tilde{Y}_t, \tag{5}$$

where

$$D(S_t) = 1 - \exp\left[-\zeta(S_t - S_0)\right], \tag{6}$$

with S_0 denoting the pre-industrial atmospheric CO_2 concentration and $\zeta > 0$ the damage parameter.

Intermediate Goods Each intermediate good, Y_t^k, $k \in \{E, H\}$, is produced competitively using a sector-specific clean input, Y_{ct}^k, and a common dirty input, Y_{dt}^k. The clean inputs represent renewable, non-emitting resources used to produce good k, while the dirty input represents GHG emitting production resources, such as fossil fuels. Sector-specificity of clean inputs accounts for differences in decarbonization pathways: electricity can be cleanly generated using solar or wind power, while green cement production

may rely on innovations such as alternative low-carbon binders (see Section A for an extensive discussion). The production function of each intermediate good is:

$$Y_t^k = \left((Y_{ct}^k)^{(\varepsilon-1)/\varepsilon} + (Y_{dt}^k)^{(\varepsilon-1)/\varepsilon} \right)^{\varepsilon/(\varepsilon-1)} \tag{7}$$

I follow widely used estimates of the elasticity of substitution between fossil fuels and renewables and assume that inputs are substitutes, $\varepsilon > 1$ (Papageorgiou et al., 2017).

Input Production The inputs are produced using labor and a continuum of sector-specific machines, according to:

$$Y_{ct}^k = (L_{ct}^k)^{1-\alpha} \int_0^1 (A_{cit}^k)^{1-\alpha} (x_{cit}^k)^{\alpha} di, \quad k \in \{E, H\} \tag{8}$$

$$Y_{dt} = (L_{dt})^{1-\alpha} \int_0^1 (A_{dit})^{1-\alpha} (x_{dit})^{\alpha} di \tag{9}$$

where $\alpha \in (0,1)$, A_{cit}^k (A_{dit}) represents the technological level of machine i in the clean input sector of intermediate good $k \in \{E, H\}$ (or the dirty input sector) at time t, and x_{cit}^k (x_{dit}) denotes the corresponding machine. The labor market clearing condition is (total labor is inelastically supplied and normalized to 1):

$$L_{ct}^E + L_{ct}^H + L_{dt} \leq 1, \tag{10}$$

Dirty input market clearing condition is (with demand not exceeding supply also in the two clean input sectors):

$$Y_{dt}^E + Y_{dt}^H \leq Y_{dt} \tag{11}$$

and final good market clearing condition is:

$$C_t = Y_t - \psi \left(\sum_{k \in \{E,H\}} \left(\int_0^1 x_{cit}^k di \right) + \int_0^1 x_{dit} di \right) \tag{12}$$

The last condition indicates that machines are supplied by monopolistically competitive firms, with each firm producing one machine at a constant exogenous cost $\psi > 0$, normalized to $\psi = \alpha^2$.

Innovation Possibilities Frontier Innovation evolves as follows. In each period, scientists choose to innovate in either one of the clean input sectors or in the dirty input sector. Assigned randomly to a machine in their chosen sector, scientists improve machine productivity by a factor of $(1 + \gamma)$ with probabilities η_c^E, η_c^H, or $\eta_d \in (0, 1)$. Successful improvements become productive in the following period. Clean technologies evolve as:

$$A_{cit}^k = \begin{cases} (1+\gamma)A_{cit-1}^k, & \text{w.p.} \quad \eta_c^k \\ A_{cit-1}^k, & \text{w.p.} \quad 1-\eta_c^k \end{cases}, \quad k \in \{E, H\} \tag{13}$$

for machines assigned a scientist in period $t-1$, while machines without a scientist retain their productivity ($A_{cit}^k = A_{cit-1}^k$). Dirty technology evolves analogously:

$$A_{dit} = \begin{cases} (1+\gamma)A_{dit-1}, & \text{w.p.} \quad \eta_d \\ A_{dit-1}, & \text{w.p.} \quad 1-\eta_d \end{cases} \tag{14}$$

Successful scientists in the clean or dirty input sectors receive a payment that depends on the number of scientists innovating in that sector. Let π_{cit}^k denote the profit of the producer of machine x_{cit}^k at time t (profits from innovation in the dirty sector, π_{dit}, are defined analogously). If a scientist is assigned to this machine and successfully improves its technology, the machine producer pays the scientist $(1 + \Gamma(s_{ct}^k))\pi_{cit}^k$. $\Gamma(\cdot)$ is a decreasing function of s_{ct}^k, non-negative on $[0, 1]$, and reflects that as more scientists work in a sector, the expected pay for each individual scientist decreases due to competition. Similar to labor, the total supply of scientists, s_t, is inelastic and normalized to 1. The market clearing condition for scientists is:

$$s_{ct}^E + s_{ct}^H + s_{dt} \leq 1 \tag{15}$$

The average sector-specific technology levels are defined as:

$$A_{ct}^k \equiv \int_0^1 A_{cit}^k di \quad k \in \{E, H\}, \qquad A_{dt} \equiv \int_0^1 A_{dit} di \tag{16}$$

which is useful for a discussion of the evolution of input technology over time. Based on the process described above, technology evolves following:

$$A_{ct}^k = \left(1 + \gamma \eta_c^k s_{ct-1}^k\right) A_{ct-1}^k, \quad k \in \{E, H\} \tag{17}$$

$$A_{dt} = (1 + \gamma \eta_d s_{dt-1})\, A_{dt-1} \tag{18}$$

That is, technology has a *building on the shoulders of giants* structure, where advancements depend on past levels of technology.

The assumptions on the evolution of technology and the compensation of scientists capture the market failure of innovation. A successful improvement permanently raises the technology level on which all subsequent innovation in the sector builds, yet the scientist's payment, $(1 + \Gamma(s^k_{ct}))\pi^k_{cit}$, reflects only the current profits from successfully improving the quality of machine x^k_{cit}. The equilibrium allocation of scientists therefore does not internalize the contribution of current innovation to future technological progress, leaving this knowledge externality unpriced.

Sector Heterogeneity in Abatement Cost To capture the greater challenge of transitioning from the dirty to the clean input in the hard-to-abate sector, the initial level of the clean technology in the hard-to-abate sector $k = H$ is assumed to be less advanced than in the easy-to-abate sector $k = E$, relative to the dirty technology:

$$\frac{A^E_{c0}}{A_{d0}} > \frac{A^H_{c0}}{A_{d0}} \iff A^E_{c0} > A^H_{c0}. \tag{19}$$

This reflects the initial underdevelopment of the clean hard-to-abate technology, while allowing it to catch up if innovation is directed toward this input sector.

The Carbon Cycle I capture environmental damages by tracking the stock of carbon in the atmosphere, S_t. The use of the dirty input generates greenhouse gas emissions and increases this carbon stock. Let emissions at date t be given by

$$P_t = \xi Y_{dt}, \tag{20}$$

where $\xi > 0$ denotes the emissions intensity of the dirty input. Following Golosov et al. (2014), I model the evolution of atmospheric carbon using the reduced-form carbon cycle specification

$$S_t = S_0 + \sum_{s=0}^{t} \left[\phi_L + (1 - \phi_L)\phi_0(1 - \phi_d)^s\right] P_{t-s}, \tag{21}$$

where S_0 denotes the pre-industrial atmospheric CO_2 concentration. In this formulation, a fraction ϕ_L of emissions remains in the atmosphere permanently, while of the remaining emissions, a fraction $1 - \phi_0$ is absorbed immediately and the rest decays geometrically at rate ϕ_d.

Climate Policy The carbon market is modeled as a cap-and-trade system with two components: an economy-wide cap on emissions (corresponding to the dirty input) and trading of emission allowances.[17] Each intermediate good firm, $k \in \{E, H\}$, receives a quota of dirty input, $\bar{Y}_{dt}^k$, that it can use in each period. This quota corresponds to emission allowances. Firms that exceed this cap must purchase additional allowances from firms using less dirty input than their allocated quota, thereby creating a market for carbon. Consequently, in each intermediate goods sector, $k \in \{E, H\}$, firms are subject to the following emission constraint:

$$Y_{dt}^k \leq \bar{Y}_{dt}^k + Z_t^k, \quad k \in \{E, H\} \tag{22}$$

where Z_t^k represents the amount of emission allowances purchased by producers of intermediate good k from producers of intermediate good $-k$ at time t ($Z_t^k < 0$ indicates that this sector is a net seller of emission allowances). The carbon market clearing condition is:

$$Z_t^E + Z_t^H = 0 \tag{23}$$

which implies that the supply of allowances equals their demand.

Each period, intermediate good producers choose input quantities Y_{ct}^k and Y_{dt}^k, and demand for emission allowances Z_t^k to maximize profits, given input prices p_{ct}^k, p_{dt}, and carbon price p_{zt}:

$$\max_{Y_{ct}^k, Y_{dt}^k, Z_t^k} \left\{ p_t^k Y_t^k - p_{ct}^k Y_{ct}^k - p_{dt} Y_{dt}^k - p_{zt} Z_t^k \right\} \quad s.t. \quad Y_{dt}^k \leq \bar{Y}_{dt}^k + Z_t^k, \quad k \in \{E, H\} \tag{24}$$

Finally, reflecting the structure of carbon markets, the cap on emissions is designed to decrease over time:

$$\bar{Y}_{dt}^k = (1 - \theta) \bar{Y}_{dt-1}^k, \quad k \in \{E, H\} \tag{25}$$

where $\theta \in (0, 1)$ represents the rate of decrease in the emissions cap. This decreasing cap

[17] Equation 20 demonstrates that an emissions cap and a cap on the use of the "dirty" input are equivalent, since emissions are the use of the "dirty" input Y_{dt} scaled by the emissions rate ξ.

aligns with the structure of the EU-ETS, which progressively tightens emissions limits year by year.

5.2 Static Equilibrium

This subsection characterizes the static equilibrium, and Section 5.3 discusses the intertemporal dynamics of the model, which are generated by the equilibrium in the market for scientists.

Each period $t = 1, ..., \infty$, an equilibrium is a sequence of (for each $k \in \{E, H\}$): (i) wages (w_t), (ii) machine prices (p^k_{cit}, p_{dit}), (iii) input prices (p^k_{ct}, p_{dt}), (iv) intermediate good prices (p^k_t), (v) demands for machines (x^k_{cit}, x_{dit}), (vi) demands for inputs (Y^k_{jt}), (vii) demands for intermediate goods (Y^k_t), (viii) demands for labor (L^k_{ct}, L_{dt}), (ix) allocations of scientists (s^k_{ct}, s_{dt}), (x) carbon prices (p_{zt}), (xi) demands for carbon allowances (Z^k_t), (xii) policy instruments ($\bar{Y}^k_{d0}$, θ) and (xiii) the atmospheric carbon concentration (S_t), such that:

- For each intermediate good sector $k \in \{E, H\}$:
  - (p^k_{cit}, x^k_{cit}) maximizes profits of the producer of clean machine i
  - (L^k_{ct}) maximizes the profits of clean input producers, given wages (w_t)
  - (Y^k_{ct}, Y^k_{dt} and Z^k_t) maximizes the profits of the intermediate good producers subject to policy constraints, given prices (p^k_{ct}, p_{dt}, p_{zt})
  - (p^E_t, p^H_t) clears the intermediate goods markets
- (p_{dit}, x_{dit}) maximizes profits of the producer of dirty machine i
- (L_{dt}) maximizes the profits of dirty input producers, given wages (w_t)
- (s^E_{ct},s^H_{ct}, s_{dt}) maximizes scientists' expected payoffs
- (Y^E_t, Y^H_t) maximizes final good producers' profits, given prices (p^E_t, p^H_t)
- Final good market clears:

$$C_t = Y_t - \psi \left(\sum_{k \in \{E,H\}} \left(\int_0^1 x^k_{cit} di \right) + \int_0^1 x_{dit} di \right) \tag{26}$$

- (w_t) clears the labor market: $L^E_{ct} + L^H_{ct} + L_{dt} \leq 1$

- The market for scientists clears: $s_{ct}^E + s_{ct}^H + s_{dt} \leq 1$
- The market for the dirty input clears: $Y_{dt} = Y_{dt}^E + Y_{dt}^H$
- (p_{zt}) clears the carbon market: $Z_t^E + Z_t^H = 0$

Appendix C.1 shows that the only two feasible equilibrium cases are (1) no carbon trading in equilibrium (none of the sectors' emissions constraints (Equation 22) bind) and (2) carbon trading in equilibrium (both of the sectors' emissions constraints bind). Appendix C.2 outlines the equilibrium solution.

5.3 Inter-temporal Equilibrium Dynamics

Finally, we turn to the decision of scientists, which determines the inter-temporal dynamics of input production technology. Given the structure of scientists' pay, the evolution of technology, equilibrium machine producer profits (Equation 49 in Appendix C.2), and the equilibrium supply of inputs (Equation 51 in Appendix C.2), the expected profits from innovating in each of the three sectors can be defined as:

$$\Pi_{ct}^k = \eta_c^k(1 + \Gamma(s_{ct}^k))\alpha(1-\alpha)A_{ct}^k L_{ct}^k (p_{ct}^k)^{\frac{1}{1-\alpha}}, \quad k \in \{E, H\} \tag{27}$$

$$\Pi_{dt} = \eta_d(1 + \Gamma(s_{dt}))\alpha(1-\alpha)A_{dt} L_{dt} p_{dt}^{\frac{1}{1-\alpha}} \tag{28}$$

which simplify to:

$$\Pi_{ct}^k = \eta_c^k(1 + \Gamma(s_{ct}^k))\alpha(1-\alpha)p_{ct}^k Y_{ct}^k, \quad k \in \{E, H\} \tag{29}$$

$$\Pi_{dt} = \eta_d(1 + \Gamma(s_{dt}))\alpha(1-\alpha)p_{dt} Y_{dt} \tag{30}$$

Scientists' expected profits in each sector increase with the beginning-of-period average technological levels A_{ct}^k and A_{dt} (*building on the shoulders of giants effect*), the price of the input, p_{ct}^k or p_{dt} (*price effect*), and the demand for the input (*market-size effect*), Y_{ct}^k or Y_{dt}. These properties align with the standard features of directed technical change (as in Acemoglu et al. (2012)). At the same time, expected profits decrease with the number of scientists innovating in a sector, introducing a *congestion effect*. To capture this, I assume $\Gamma(s) = \frac{\gamma}{s}$ for $s \in \{s_{ct}^E, s_{ct}^H, s_{dt}\}$, where γ is the innovation step size. This formulation results in a unique interior equilibrium for the share of scientists in each sector, such that $s_{ct}^E, s_{ct}^H, s_{dt} \in (0, 1)$ for each $t = 1, ..., \infty$, and allocations of scientists equalize expected pay

across sectors:

$$\Pi_{ct}^{E}(s_{ct}^{E}) = \Pi_{ct}^{H}(s_{ct}^{H}) = \Pi_{dt}(s_{dt}) \tag{31}$$

Appendix C.3 provides a proof of this result.

5.4 Research Subsidies

The directed technical change literature has shown how carbon taxation and subsidies to clean research interact to lead to an efficient energy transition (Acemoglu et al., 2012, 2016, 2023). I therefore analyze the interaction between a carbon market and *sector-specific* subsidies to clean research. A subsidy of size $q_t^k > 0$ increases scientists' expected profits from innovation in clean input sector $k \in \{E, H\}$ at time t by a factor of $1 + q_t^k$:

$$\Pi_{ct}^{k} = (1 + q_t^k)\eta_c^k(1 + \Gamma(s_{ct}^k))\alpha(1-\alpha)A_{ct}^k L_{ct}^k (p_{ct}^k)^{\frac{1}{1-\alpha}}, \quad k \in \{E, H\} \tag{32}$$

As discussed above regarding the equilibrium allocation of scientists, subsidies to clean research increase the equilibrium share of scientists innovating in the clean sector by providing a markup to their expected profits. These subsidies are financed by the household.

6 Model Calibration

This section presents the results from a calibration of the model. The purpose of the calibration is to provide theoretical foundations for the empirical findings on the differential effect of the EU-ETS on innovation activity and to shed light on their implications for the efficiency of the energy transition under a carbon market. After outlining the setup and specifying the parameter values, I present the calibration results, and finish by presenting several robustness checks demonstrating that the qualitative findings do not hinge on specific parameter choices.

6.1 Setup

Optimal Policy Problem The optimal policy is defined as a sequence of subsidies $q_t \equiv \{q_t^E, q_t^H\}$ and an emissions cap decrease rate θ that maximize household utility, subject to the final good market clearing condition. Thus, the optimal policy can be characterized

as the solution to the following planning problem:

$$\max_{\{q_t\}_{t=1}^{T},\theta} \sum_{t=1}^{T} \frac{1}{(1+\rho)^t} u\left(C_t(q_t,\theta)\right) \tag{33}$$

s.t.

$$C_t(q_t,\theta) \leq \underbrace{Y_t(q_t,\theta)}_{\text{Final good}} - \underbrace{\psi\left(\sum_{k\in\{E,H\}}\left(\int_0^1 x_{cit}^k(q_t,\theta)di\right)+\int_0^1 x_{dit}(q_t,\theta)di\right)}_{\text{Cost of machines}} - \underbrace{Q_t(q_t,\theta)}_{\text{Subsidies cost}}$$

The results focus on the effect of subsidies on the transition to clean production. Therefore, the cap decrease rate θ is taken as given, so we can contrast the transition dynamics under (1) a no subsidies scenario ($q_t^H = q_t^E = 0, t = 1, 2, ..., T$) and (2) an optimal subsidies scenario ($\{q_t^H, q_t^E\}$ are the solution to the planning problem outlined above, given θ). $Q_t(\cdot)$ is the total amount paid in subsidies to scientists innovating in either clean input sectors, and is equal to:

$$Q_t(q_t,\theta) = \sum_{k\in\{E,H\}} q_t^k \eta_c^k s_{ct}^k (1+\Gamma(s_{ct}^k))\alpha(1-\alpha)A_{ct}^k L_{ct}^k (p_{ct}^k)^{\frac{1}{1-\alpha}} \tag{34}$$

As is common in the literature, the model is calibrated for an economy with $T = 400$ years time horizon, which provides a long horizon to allow (potential) climate damages to realize.[18]

Parameter Values Table 7 summarizes the parameter values used in the calibration. The share of machines is set to match the share of capital in the economy, $\alpha = 1/3$. The elasticity of substitution between inputs is set to $\varepsilon = 2.46$ following Papageorgiou et al. (2017), and the elasticity of substitution between intermediate goods is proxied using estimates of top-level elasticity in nested CES production functions (Van der Werf, 2008), and is set to $\nu = 0.36$.[19] Following Golosov et al. (2014) and Acemoglu et al. (2023), the

[18] Acemoglu et al. (2012) and Acemoglu et al. (2023) consider 400 years as well and Cai and Lontzek (2019) use 600 years.

[19] For ε, I use the average CES-based estimate of the elasticity of substitution between clean and dirty inputs across electricity and non-energy sectors. For ν, I use the cross-country average of the estimated top-level elasticity of substitution between energy and production inputs. Section 6.3 confirms that the results are robust to variations in these parameters.

utility function is assumed to take a CRRA form:

$$u\left(C_t\right) = \frac{C_t^{1-\sigma}}{1-\sigma} \tag{35}$$

The utility function is specified with a pure rate of time preference $\rho = 1\%$ and a coefficient of relative risk aversion $\sigma = 1.5$, following the 2023 version of the Dynamic Integrated Climate-Economy (DICE) model (Barrage and Nordhaus, 2023). Climate change affects welfare through the damage function $D(S_t)$, which reduces final output as specified below.

The innovation step size is set to $\gamma = 1$ without loss of generality, and the probabilities of innovation success are set to $\eta_d = 0.02$, $\eta_c^E = 0.04$, and $\eta_c^H = 0.04$ to match a 2% growth rate in the balanced growth path (BGP).[20] The assumption of equal success probabilities of innovation in the two clean input sectors, $\eta_c^E = \eta_c^H = 0.04$, implies that although the clean hard-to-abate technology starts at a disadvantage, once the sector catches up and technological levels converge, the expected innovation step size will also equalize at $\eta_c^E\gamma = \eta_c^H\gamma$. Due to the convex nature of the innovation process (*building on the shoulders of giants*), the clean hard-to-abate technology will initially advance with smaller absolute gains.

I use the International Energy Agency's (IEA) World Energy Statistics and Balances to determine the initial usage of fossil fuels and renewable energy by each sector, Y_{d0}^k and Y_{c0}^k (International Energy Agency, 2024). These data provide detailed information on the use of over 60 energy sources (e.g., coal, crude oil, nuclear energy, and renewable energy) by nearly 100 intermediate- (e.g., electricity production) and end-users of energy (disaggregated industrial sectors, as well as residential, commercial, and transportation sectors), covering more than 150 countries since 1971. This information allows me to determine the initial input values for each intermediate good sector and to derive initial technological levels based on equilibrium conditions.

In an effort to capture the global economy, the easy-to-abate sector includes light industry (transport equipment, machinery, food and tobacco, wood and wood products, textile and leather), rail transportation, and residential and commercial uses of energy.

[20] In the decentralized equilibrium, without climate policy in place, all scientists innovate in the dirty sector, resulting in a BGP growth rate of $\eta_d\gamma$. In the equilibrium with a carbon market, the emission cap eventually becomes infinitely stringent, $\lim_{t\to\infty} \bar{Y}_{dt} = 0$, causing the number of scientists in each clean sector to converge to 0.5, $\lim_{t\to\infty} s_{ct}^k = 0.5, k \in \{E, H\}$. This implies a BGP growth rate of $0.5(0.5\eta_c^E\gamma + 0.5\eta_c^H\gamma)$.

The hard-to-abate sector is defined to include mining and quarrying, construction, iron and steel, chemical and petrochemical, non-ferrous metals, non-metallic minerals, paper, pulp and printing, aviation, road transport, and maritime shipping (Industry titles as they appear in the data. See Sections 2.2 and Appendix A for further details).

Initial values of input use in each sector are set to the total amount of fossil fuels and renewable energy consumed by each sector,[21] including indirect energy use via electricity, during the period 2018-2021.[22]

The focus on end-users of energy rather than primary users ensures consistency with the interpretation of intermediate goods as consumption goods and avoids double counting of energy used for electricity. Including electricity as part of the easy-to-abate sector would double count its energy consumption, once for energy used in electricity generation and again for energy consumed by hard-to-abate industries. Additionally, this approach allows me to include the residential and commercial sectors, which primarily derive their clean energy consumption indirectly through renewable electricity used for heating, cooling, and air conditioning. Thus, I adopt the end-user approach, including both direct and indirect energy use.

The initial values of dirty and clean inputs for the easy- and hard-to-abate sectors are $Y_{c0}^E = 268,681,907$, $Y_{d0}^E = 550,475,222$, $Y_{c0}^H = 86,192,545$ and $Y_{d0}^H = 776,408,740$ Terajoules. These values indicate that fossil fuels account for 90% of energy consumption in the hard-to-abate sector and 67.2% in the easy-to-abate sector.

Initial input values subsequently determine the initial technology levels for the production of each input, based on equilibrium conditions. The implied technological gaps are:

$$\frac{A_{d0}}{A_{c0}^E} = 1.73, \quad \frac{A_{d0}}{A_{c0}^H} = 3.47 \tag{36}$$

These gaps imply that the easy-to-abate sector starts with a more advanced clean technology relative to the hard-to-abate sector, making it comparatively cheaper to produce using clean inputs in the easy-to-abate sector. Put simply, the initial clean easy-to-abate technology is twice as advanced as the initial clean hard-to-abate technology.

[21] Fossil fuels include Coal and Coal Products, Peat and Peat Products, Oil Shale and Oil Sands, Crude, NGL (Natural Gas Liquids) and Feedstocks, Oil Products, and Natural Gas. Renewable energy includes Nuclear, Hydro, Geothermal, Solar/Wind/Other, Biofuels, and Waste (names are as provided in the IEA data).

[22] A four-year period is used because the model is discretized into four-year intervals for computational reasons.

I rely on the initial values of dirty-input usage and observed baseline emissions to calibrate the emissions intensity of the dirty input, ξ. Let total emissions in period t be given by

$$P_t = \xi Y_{dt}, \tag{37}$$

where Y_{dt} denotes aggregate dirty-input use and $\xi > 0$ is the associated emissions intensity. The parameter ξ is pinned down by the initial emissions-output ratio,

$$\xi = \frac{P_0}{Y_{d0}}, \tag{38}$$

where P_0 denotes baseline gross CO_2 emissions over one four-year period (2018–2021), expressed in ppm, and $Y_{d0} = Y_{d0}^H + Y_{d0}^E$ denotes baseline dirty-input use, expressed in quadrillion Btu.[23] In the quantitative implementation, this implies

$$\xi = \frac{16.9719}{1257.6433} \approx 0.0135. \tag{39}$$

Following Golosov et al. (2014), I model the evolution of atmospheric carbon using a reduced-form carbon cycle in which a fraction of emissions remains permanently in the atmosphere, while the remainder is either absorbed immediately or decays gradually over time. Following Acemoglu et al. (2023), I adapt this specification to the model's four-year time periods by setting $\phi_L = 0.2$ and choosing the remaining parameters to match the same key moments of carbon persistence. In particular, the decay parameter ϕ_d is chosen to match a 300-year half-life of the slowly decaying carbon component, implying

$$\phi_d = 1 - 0.5^{4/300} \approx 0.0092. \tag{40}$$

The parameter ϕ_0 is then chosen so that approximately half of a CO_2 impulse is removed after 30 years, which in the present setting implies

$$\phi_0 = \frac{0.5 - \phi_L}{(1 - \phi_L)(1 - \phi_d)^{30/4-1}} \approx 0.3982. \tag{41}$$

Let S_t denote atmospheric CO_2 concentration and let S_0 denote its pre-industrial level.

[23] Global CO_2 emissions over 2018–2021 total 132.7 GtCO2, converted to ppm at 7.82 GtCO2 per ppm. Dirty-input use is converted from Terajoules at 1 quadrillion Btu = 1,055,055.9 Terajoules.

The stock of atmospheric carbon is given by

$$S_t = S_0 + \sum_{s=0}^{t} \left[\phi_L + (1-\phi_L)\phi_0(1-\phi_d)^s\right] P_{t-s}. \tag{42}$$

This formulation reflects that a fraction ϕ_L of emissions remains permanently in the atmosphere, while of the remaining emissions, a fraction $1-\phi_0$ is absorbed immediately and the rest decays geometrically at rate ϕ_d.

I also adopt the exponential damage specification of Golosov et al. (2014), as in Acemoglu et al. (2023), so that the fraction of output lost due to atmospheric carbon concentration is

$$D(S_t) = 1 - \exp\left[-\zeta(S_t - S_0)\right], \tag{43}$$

where $\zeta > 0$ is the damage parameter. I use the benchmark value $\zeta = 5.3 \times 10^{-5}$ from Golosov et al. (2014). Since atmospheric carbon is measured in ppm in the quantitative implementation, this parameter is rescaled using the conversion factor 1 ppm $\approx$ 2.13 GtC. The emissions cap decrease rate is set to $\theta = 5.25\%$, which achieves climate neutrality by 2050 in the model.[24] Robustness checks confirm that the results hold for less stringent climate policy (i.e., lower emissions cap decrease rate θ).

6.2 Calibration Results

This section presents calibration results and compares transition dynamics under (1) a carbon market consistent with climate neutrality by 2050 ambitions (the no-subsidies scenario) and (2) the same carbon market supplemented by optimal research subsidies (the optimal-subsidies scenario). In both cases, the stringency of climate policy is held constant via the same emissions cap decrease rate, θ. The comparison highlights the critical role of subsidies in facilitating a transition that achieves higher welfare by re-aligning incentives to increase clean innovation in the hard-to-abate sector.

Figure 5 presents the path of optimal subsidies to scientists innovating in the clean input of the easy- and hard-to-abate sectors, q^E and q^H. The social planner allocates much higher subsidies to scientists innovating in the hard-to-abate sector (170% higher in present value). As shown below, carbon trading prolongs the dependency of hard-to-

[24] A decrease rate of $\theta = 5.25\%$ corresponds to the average yearly emissions decrease from 2021 global GHG emissions (36.3 Gigatons of GHG emissions, International Energy Agency (2022)) to the IEA's Net Zero by 2050 prediction for CO_2 capture levels in 2050 (7.6 Gigatons of GHG emissions will be captured and not abated, International Energy Agency (2021)).

abate producers on fossil fuels Y_{dt}^H, depressing the demand for the clean hard-to-abate input Y_{ct}^H, and reducing the expected payoff from innovating in that sector Π_{ct}^H, thereby widening the gap between the social and private return to clean innovation there. Subsidies counteract this effect, and are used to draw scientists to the clean hard-to-abate input sector. Furthermore, as in Acemoglu et al. (2012), optimal subsidies are short-lived and phase out completely by 2050 and 2070 in the easy- and hard-to-abate sectors, respectively.

Figure 6 compares the allocation of scientists (s_{ct}^E, s_{ct}^H, s_{dt}) across the two scenarios. With optimal subsidies (Panel a), most scientists are immediately drawn to the hard-to-abate sector, increasing s_{ct}^H. Without subsidies (Panel b), scientists initially concentrate in the easy-to-abate sector, where carbon markets accelerate decarbonization, boosting demand for clean inputs Y_{ct}^E and raising expected payoffs Π_{ct}^E via market size and price effects.

Figure 7 compares the technological gap between the dirty and the clean input in each sector (A_{dt}/A_{ct}^k, $k \in \{E, H\}$). Without subsidies, the gap in the hard-to-abate sector widens at first (Panel a). In contrast, optimal subsidies immediately shrink the gap by increasing clean innovation in that sector. The role of subsidies in the easy-to-abate sector is marginal, and the dynamics of the technological gap under the two scenarios are very similar (Panel b).

Figure 8 presents the paths of emission allowances purchased by hard-to-abate firms, Z_t^H, under the no-subsidies and optimal-subsidies scenarios. The figure demonstrates the significant impact of subsidies on carbon trading. As subsidies align innovation incentives and accelerate advancements in clean hard-to-abate technology A_{ct}^H, total carbon trading volume shrinks by 75%, and the use of allowances Z_t^H is phased out decades earlier. This underscores the inefficiency of allowing hard-to-abate firms to rely heavily on emission allowances instead of undertaking direct emissions reduction. In the absence of targeted subsidies, carbon markets alone lead to excessive carbon trading, delaying technological progress and undermining the efficiency of the transition.

Figure 9 compares production in the easy- and hard-to-abate sectors, Y_t^E and Y_t^H, under both scenarios. Enforcing the emission reduction pace required to reach climate neutrality by 2050 through a carbon market causes an initial decline in intermediate good production and overall economic activity. Since clean technologies are initially underdeveloped, abatement combines technological transition, reduced production, and, solely in the hard-to-abate sector, purchase of emission allowances. Targeted research subsidies

soften this initial contraction in the hard-to-abate sector (Y_t^H) and accelerate its recovery (Panel a), while having a smaller long-run effect in the easy-to-abate sector, where the carbon market already delivers strong clean innovation incentives (Panel b).

Finally, the re-direction of innovation has a significant impact on household consumption C_t. Figure 10 shows that the initial decline in consumption under the no-subsidies scenario (due to the stringency of the climate policy required to achieve net-zero emissions by 2050) is largely mitigated with optimally designed subsidies. In consumption equivalent terms, subsidies increase welfare by 3.8%, which is a result of avoiding the contraction in the production of the hard-to-abate intermediate good Y_t^H. The welfare gain speaks to the cost of a knowledge externality that carbon trading exacerbates: allowance purchases depress the private return to clean innovation in the hard-to-abate sector while its social return remains high, and the differentially higher subsidies price this widened gap. Notably, conditional on the emission cap decrease rate θ, subsidies do not affect the economy-wide pace of decarbonization, which is equivalent to the emissions cap decrease rate θ. Thus, subsidies complement the cap-and-trade system by redirecting innovation to the hard-to-abate sector, allowing for a more efficient energy transition.

6.3 Robustness Checks

The robustness checks presented below explore variations in the main parameters. Table 8 summarizes these checks, demonstrating that the baseline calibration results do not qualitatively depend on the parameter values used in the main calibration of the model (Table 7). In each check, only the specified parameter is altered, and the others are specified in Table 7.

The pure rate of time preference in the robustness check, $\rho = 0.1\%$, is selected following Stern (2007). The discount rate is often particularly important for climate policy analyses and forecasts due to the long time horizons involved, where benefits from climate policy often materialize in the distant future. The Stern pure rate of time preference incorporates a moral concern for future generations, unlike the market-based measures of Barrage and Nordhaus (2023) used in the main specification (Golosov et al., 2014). Since these discount rates often lead to differing climate policy implications, it is reassuring to confirm that the results remain consistent under both assumptions.

Additionally, I vary the elasticity of substitution between intermediate goods, ν, ensuring that the results hold for varying levels of complementarity between the goods. For the elasticity of substitution between clean and dirty inputs ε, I test higher and lower val-

ues within the estimation range provided by Papageorgiou et al. (2017). Finally, I assess the robustness of the results under a lower emissions cap decrease rate ($\theta = 2.2\%$), which matches the 2021-2023 decrease rate of the EU-ETS. These checks confirm that carbon markets exhibit the side effect identified in this paper also under less stringent climate policy conditions.

7 Conclusions

This paper examines the unintended consequences of carbon markets and their implications for the efficiency of the transition to clean technology. The empirical analysis demonstrates that the EU-ETS spurred green patenting in easy-to-abate industries, with no comparable increase in hard-to-abate industries. Additionally, emissions reduction in hard-to-abate installations occurred at only about a quarter of the annual pace of their easy-to-abate counterparts.

A directed technical change model of a carbon market and sectoral differences in abatement costs sheds light on the mechanisms behind these disparities and shows that the empirical findings point to an inefficiency of carbon markets. While carbon markets effectively accelerate decarbonization in the easy-to-abate sector, they inadvertently enable hard-to-abate producers to delay independent emissions reductions by relying on the purchase of emission allowances. These dynamics disincentivize clean innovation in the hard-to-abate sector, exacerbating the knowledge externality that already leaves innovation underprovided. The model demonstrates that rebalancing innovation incentives through sector-specific research subsidies for clean innovation in the hard-to-abate sector can accelerate its transition, mitigate economic contraction, and generate welfare gains of 3.8%. However, descriptive evidence suggests that current public funding is heavily skewed, with hard-to-abate industries receiving only 7% of the financial support per ton of emissions that easy-to-abate industries receive (Table 1).

As the climate neutrality by 2050 target approaches and adoption of carbon markets grows, it is crucial to recognize and address the trade-off inherent in these systems between static cost-effectiveness and dynamic efficiency. Rather than further strengthening incentives for the decarbonization pathways in easy-to-abate industries, policymakers should adopt complementary measures to ensure a more balanced transition. Specifically, subsidies and public financing should support innovation required for the decarbonization of hard-to-abate industries, allowing for a more efficient technological transition.

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Tables

Table 1: Flows of Public Climate Finance

	Billions of $		$ Invested per tCO_2 Emitted	
	2019-2022	**2022**	**2019-2022**	**2022**
Energy Systems	728	292	12.8	19.9
Industry	30.8	14.4	0.9	1.6

Note: This table presents flows of climate finance from public institutions toward the industrial sector and energy systems in billions of USD and as $1 invested per 1 ton of CO_2 emitted. Emissions data are from International Energy Agency (2023*a*) and investment data are from the Global Landscape of Climate Finance (GLCF) 2023, collected by the Climate Policy Initiative. Public sources include development finance institutions (DFIs), governments and climate funds. Industry includes industrial, extraction and manufacturing processes, and industry infrastructure, and Energy Systems includes power and heat generation and transmission and fuel production. For more details, see GLCF 2023 methodology.

Table 2: Number of Firms by Country

Country	Non EU-ETS	EU-ETS	Total
Austria	37	11	48
Belgium	42	21	63
France	224	58	282
Germany	560	113	673
Italy	285	45	330
Netherlands	44	12	56
Spain	91	29	120
Sweden	43	19	62
United Kingdom	145	38	183
Other Countries	109	43	152
Total	1580	389	1969

Note: This table presents the number of firms by country in the main sample constructed in Section 3.2 for the 9 largest countries that participate in the EU-ETS. The rest of the countries are aggregated under "Other Countries".

Table 3: Number of Firms by Industry

Industry	Non EU-ETS	EU-ETS	Total	Hard to Abate
Chemicals	473	103	576	Yes
Electricity & Power	34	18	52	No
Food, Beverages & Tobacco	172	45	217	No
Metals	161	51	212	Yes
Mining & Oil	22	18	40	Yes
Non-Metallic Minerals	169	74	243	Yes
Pulp, Paper & Cardboard	117	35	152	Yes
Trans. Manufacturing	432	45	477	No
Total	1580	389	1969	

Note: This table presents the number of firms by industry in the main sample constructed in Section 3.2. ”Chemicals” is NACE code C20, ”Electricity & Power” is D35, ”Food, Beverages & Tobacco” is C10, C11 and C12, ”Metals” is C24, ”Mining & Oil” is B and C19, ”Non-Metallic Minerals” is C23, ”Pulp, Paper & Cardboard” is C17 and ”Trans. Manufacturing” is C29 and C30.

Table 4: The Heterogeneous Effect of the EU-ETS on Green Innovation

	(1) Green patenting	(2) Green patenting	(3) Green patenting	(4) Green patenting
Post	-0.0095***	0.0045	0.00099	-
	(0.0035)	(0.0032)	(0.0054)	
ETS	0.0047	-	-	-
	(0.0098)			
Hard	0.0042	-	-	-
	(0.0042)			
Post × ETS	0.052***	0.064***	0.066***	0.069***
	(0.018)	(0.019)	(0.019)	(0.018)
Post × Hard	0.00097	-0.0093***	-0.012***	-
	(0.0039)	(0.0036)	(0.0038)	
ETS × Hard	0.064***	-	-	-
	(0.016)			
Post × ETS × Hard	-0.066***	-0.061***	-0.061***	-0.063***
	(0.020)	(0.020)	(0.020)	(0.020)
Firm FE	N	Y	Y	Y
Country × Year FE	N	N	Y	Y
Industry × Year FE	N	N	N	Y
Uncond. Mean	0.028	0.028	0.028	0.028
Observations	37,044	37,041	36,981	36,981
Adj. R-squared	0.13	0.42	0.42	0.43

Note: This table presents triple-difference regression results on engagement in green innovation. The dependent variable is an indicator variable set to 1 if the firm filed an own-technology green patent in that year, defined as a green patent in a technology relevant to the firm's own easy- or hard-to-abate activity (see Section 3.2). Post is set to 1 for observations in post EU-ETS years, ETS is set to 1 for firms that are regulated by the EU-ETS (that is, operate at least one installation that participates in the EU-ETS), and Hard is set to 1 for firms that operate in hard-to-abate industries. The sample is the main sample constructed as explained in Section 3.2. All columns control for the lagged emissions-to-free-allocation ratio and the lagged log patent stock; columns (2)-(4) include firm fixed effects, columns (3) and (4) include country-year fixed effects, and column (4) includes industry-year fixed effects. Standard errors (in parentheses) are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: The Heterogeneous Effect of the EU-ETS on Green Innovation - Split Difference in Differences

	(1) Green patenting	(2) Green patenting	(3) Green patenting	(4) Green patenting	(5) Green patenting	(6) Green patenting	(7) Green patenting	(8) Green patenting	(9) Green patenting
Post	-0.0021	0.0042	-0.0049	-0.0092**	-	-0.011	-	-	-
	(0.0021)	(0.0032)	(0.0041)	(0.0043)		(0.0067)			
Post × ETS	0.019**	0.064***	0.0035	0.019**	0.066***	0.0045	0.024***	0.067***	0.0049
	(0.0077)	(0.019)	(0.0077)	(0.0077)	(0.019)	(0.0076)	(0.0080)	(0.019)	(0.0083)
Sample	All	Easy	Hard	All	Easy	Hard	All	Easy	Hard
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country × Year FE	N	N	N	Y	Y	Y	Y	Y	Y
Industry × Year FE	N	N	N	N	N	N	Y	Y	Y
Uncond. Mean	0.028	0.025	0.029	0.028	0.025	0.029	0.028	0.025	0.029
Observations	37,041	14,030	23,011	36,981	13,985	22,927	36,981	13,985	22,927
Adj. R-squared	0.42	0.39	0.44	0.42	0.39	0.44	0.42	0.39	0.45

Note: This table presents difference-in-differences regression results on engagement in green innovation for all firms (columns (1), (4) and (7)), firms in easy-to-abate industries (columns (2), (5) and (8)) and firms in hard-to-abate industries (columns (3), (6) and (9)). The dependent variable is an indicator variable set to 1 if the firm filed an own-technology green patent in that year, defined as a green patent in a technology relevant to the firm's own easy- or hard-to-abate activity (see Section 3.2). Post is set to 1 for observations in post EU-ETS years, ETS is set to 1 for firms that are regulated by the EU-ETS (that is, operate at least one installation that participates in the EU-ETS), and Hard is set to 1 for firms that operate in hard-to-abate industries. The sample is the main sample constructed as explained in Section 3.2. The "Sample" row reports which firms each column uses, and "Uncond. Mean" is the unconditional mean of the dependent variable. All columns include firm fixed effects and control for the lagged emissions-to-free-allocation ratio and the lagged log patent stock; columns (4)-(9) include country-year fixed effects, and columns (7)-(9) include industry-year fixed effects. Standard errors (in parentheses) are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: The Heterogeneous Effect of the EU-ETS on Green Innovation - Additional Tests and Robustness Checks

	(1) Green patenting	(2) log(1+green count)	(3) log(1+green citations)	(4) Green patenting	(5) Green patenting	(6) Green patenting	(7) Green patenting	(8) Green patenting	(9) Green patenting
Post × ETS	0.069***	0.064***	0.15***	0.050***	0.049***	0.062***	0.092***	0.089***	0.024**
	(0.018)	(0.018)	(0.043)	(0.015)	(0.015)	(0.016)	(0.025)	(0.027)	(0.0099)
Post × ETS × Hard	-0.063***	-0.056***	-0.14***	-0.040**	-0.034*	-0.034*	-0.084***	-0.081***	-0.029***
	(0.020)	(0.019)	(0.047)	(0.018)	(0.018)	(0.019)	(0.027)	(0.029)	(0.011)
Specification	Baseline	Intensive margin	Quality adjusted	Broad own-tech	Green, not other-sector	Any green patent	Large economies	Large economies	Continuous abatement cost
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country × Year FE	Y	Y	Y	Y	Y	Y	Y	N	Y
Industry × Year FE	Y	Y	Y	Y	Y	Y	Y	N	Y
Ctry × Ind. × Year FE	N	N	N	N	N	N	N	Y	N
Uncond. Mean	0.028	0.024	0.059	0.033	0.035	0.043	0.030	0.029	0.024
Observations	36,981	36,981	36,981	36,981	36,981	36,981	27,505	27,129	22,465
Adj. R-squared	0.43	0.48	0.45	0.45	0.46	0.47	0.47	0.45	0.41

Note: This table presents robustness checks for the heterogeneous effect of the EU-ETS on green innovation, discussed in Section 4.1. The baseline outcome is an indicator set to 1 if the firm filed an own-technology green patent in that year, defined as a green patent in a technology relevant to the firm's own abatement category (Section 3.2). This is the dependent variable in columns (1) and (4)-(9), except that column (4) uses a coarser classification of Y02 subcategories, column (5) sets the indicator variable to one for any green patent not classified exclusively in the other sector, and column (6) sets it to one for any green patent regardless of the firm's abatement category (see Section 4.1). In column (2) the dependent variable is the natural logarithm of one plus the number of own-technology green patents filed by the firm in that year, and in column (3) the natural logarithm of one plus the number of forward citations received by those patents. Columns (7) and (8) restrict the sample to firms in France, Germany, Italy and the United Kingdom, and column (9) replaces the binary hard-to-abate indicator with a continuous measure of industry emissions intensity (CO_2 per unit of value added). Post is set to 1 for observations in post EU-ETS years, ETS is set to 1 for firms that are regulated by the EU-ETS (that is, operate at least one installation that participates in the EU-ETS), and Hard is set to 1 for firms that operate in hard-to-abate industries. The "Specification" row labels each column. All columns include firm fixed effects and control for the lagged emissions-to-free-allocation ratio and the lagged log patent stock; columns (1)-(7) and (9) include country-year and industry-year fixed effects, and column (8) includes country-industry-year fixed effects. Standard errors (in parentheses) are clustered at the firm level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 7: Calibration Setup - Parameter Values

Parameter	Definition	Value	Source
α	machine share	1/3	Capital share
ε	elasticity of substitution between clean and dirty inputs	2.46	Papageorgiou et al. (2017)
ν	elasticity of substitution between intermediate goods	0.36	Van der Werf (2008)
σ	coefficient of relative risk aversion	1.5	Barrage and Nordhaus (2023)
ρ	pure rate of time preference	1%	Barrage and Nordhaus (2023)
γ	innovation step size	1	Match 2% BGP growth rate
$\eta_d, \eta_c^H, \eta_c^E$	innovation success probabilities	0.02, 0.04, 0.04	Match 2% BGP growth rate
ξ	emissions intensity of dirty input	0.0135	Baseline gross emissions and initial dirty-input use
ϕ_L, ϕ_0, ϕ_d	carbon cycle parameters	0.20, 0.3982, 0.0092	Adapted from Golosov et al. (2014) to simulation period length
ζ	climate damage parameter	5.3×10^{-5}	Golosov et al. (2014)
θ	cap decrease rate	5.25%	Climate neutrality by 2050 targets
$\frac{A_{d0}}{A_{c0}^E}, \frac{A_{d0}}{A_{c0}^H}$	initial technology gaps	1.73, 3.47	Equilibrium conditions and IEA World Energy Balances data

Note: This table presents the parameter values used in the baseline calibration.

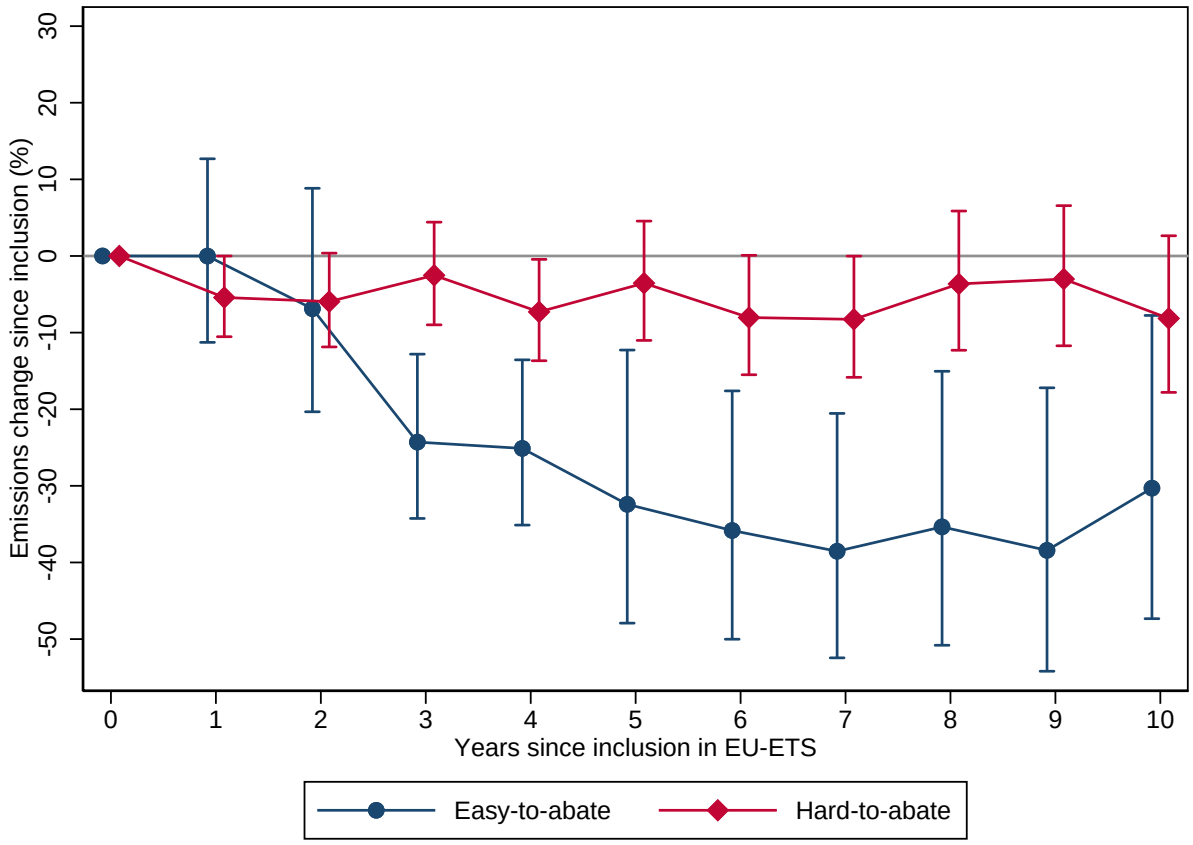
Table 8: Robustness Checks - Summary

Specification	**Param. Value**	**Ratio of PV(Subs.)**	**Welfare Effect of Subs.**	**Results**
Main	-	170%	3.8%	Fig. 5-10
Pure rate of time pref. $\downarrow$	$\rho = 0.1\%$	122%	5.8%	Fig. A.2
Subs. of interm. goods $\uparrow$	$\nu = 0.75$	37%	3.4%	Fig. A.3
Subs. of interm. goods $\downarrow$	$\nu = 0.2$	211%	3.9%	Fig. A.4
Subs. of inputs $\uparrow$	$\varepsilon = 2.87$	128%	3.8%	Fig. A.5
Subs. of inputs $\downarrow$	$\varepsilon = 2.05$	239%	3.7%	Fig. A.6
Cap decrease rate $\downarrow$	$\theta = 2.2\%$	63%	5%	Fig. A.7

Note: This table presents results from robustness checks, varying different parameters. In each check, only the parameter listed changes in value, and the rest of the parameters are set as in Table 7. "Ratio of PV(Subs.)" is the ratio of the present value of subsidies paid to scientists innovating in the hard-to-abate sector over the present value of subsidies paid to scientists innovating in the easy-to-abate sector minus 1. "Welfare Effect of Subs." is the percent increase in equivalent consumption terms comparing the optimal subsidies to the no subsidies case.

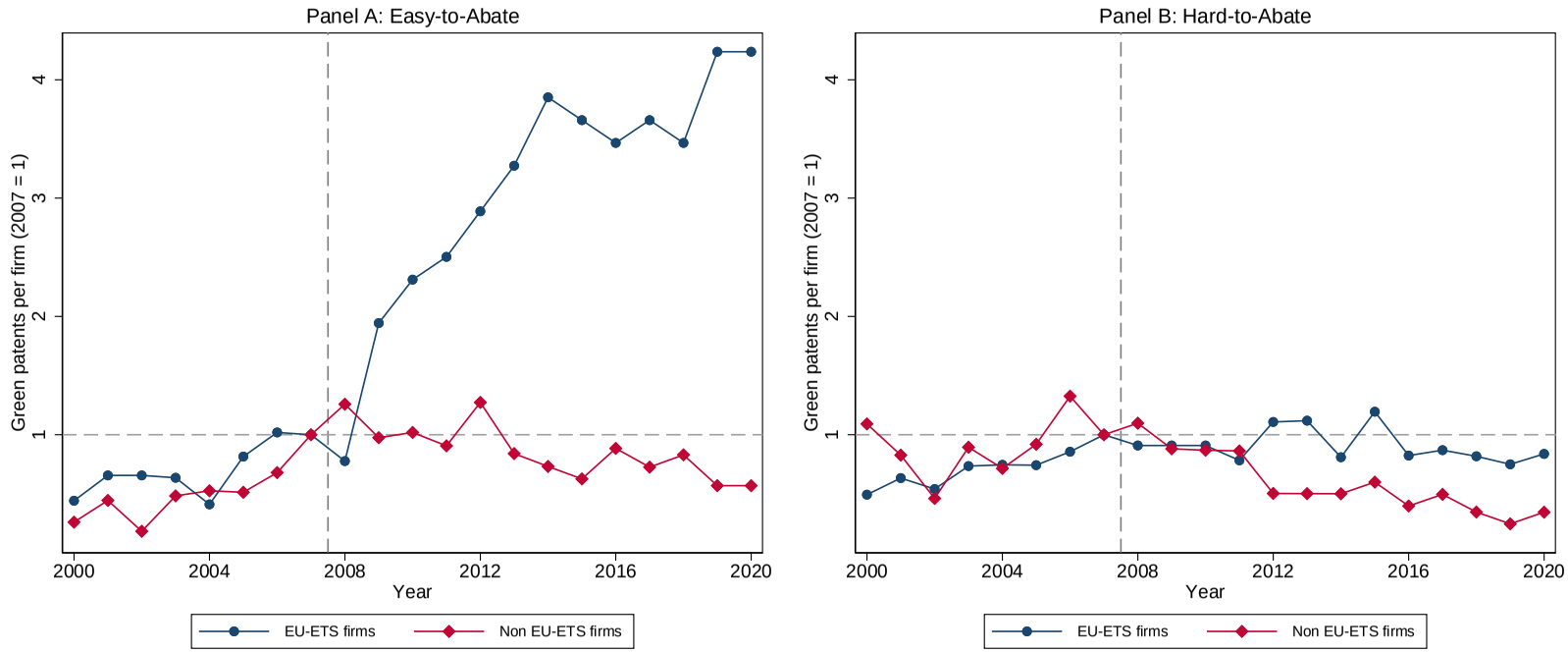
Figures

Figure 1: Change in emissions: easy- and hard-to-abate installations



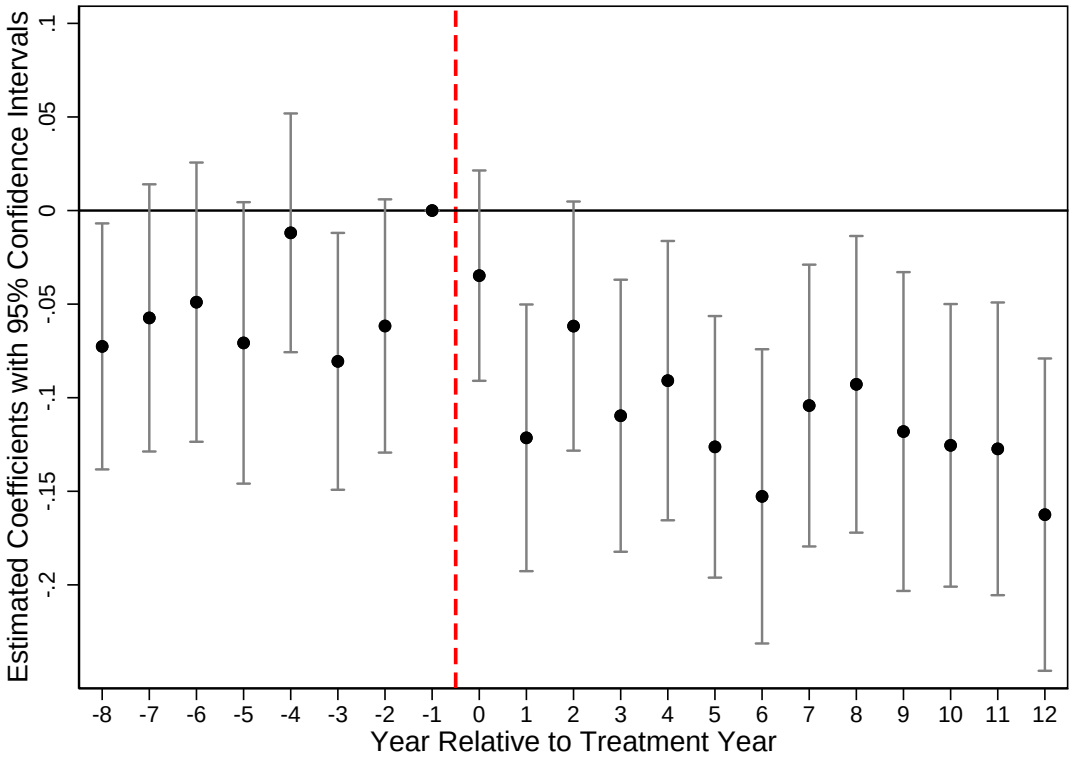
Note: This figure plots the estimated β_τ coefficients from Equation (1), estimated separately for easy-to-abate (blue, circles) and hard-to-abate (red, diamonds) EU-ETS installations. $\log(emissions_{it})$ is the natural log of the yearly emissions of installation i, α_i are installation fixed effects, and $I\{\text{event time}_{it} = \tau\}$ are indicators for the number of years τ since installation i is first observed in the EU-ETS; the omitted reference is the year of inclusion ($\tau = 0$). Each point reports the implied percentage change, $100 \cdot (\exp(\beta_\tau) - 1)$, with 95% confidence intervals obtained by exponentiating the corresponding bounds on β_τ. The sample includes all installations operated by main sample firms, as defined in Section 3.2, that participate in the EU-ETS. Standard errors are clustered at the installation level.

Figure 2: The EU-ETS and the Trend in Green Patenting of Easy- and Hard-to-abate Firms



Note: This figure presents the per firm average number of green patents filed by EU-ETS (blue, dots) and non EU-ETS (red, diamonds) firms in 2000-2020. Panel A includes firms in easy-to-abate industries and Panel B includes firms in hard-to-abate industries. The annual firm-level green patent count is winsorized at the 1st and 99th percentiles. Each series is normalized to 1 at 2007, one year prior to the full operation of the EU-ETS.

Figure 3: Dynamic Effect of EU-ETS on Green Patenting in Hard-to-Abate Industries

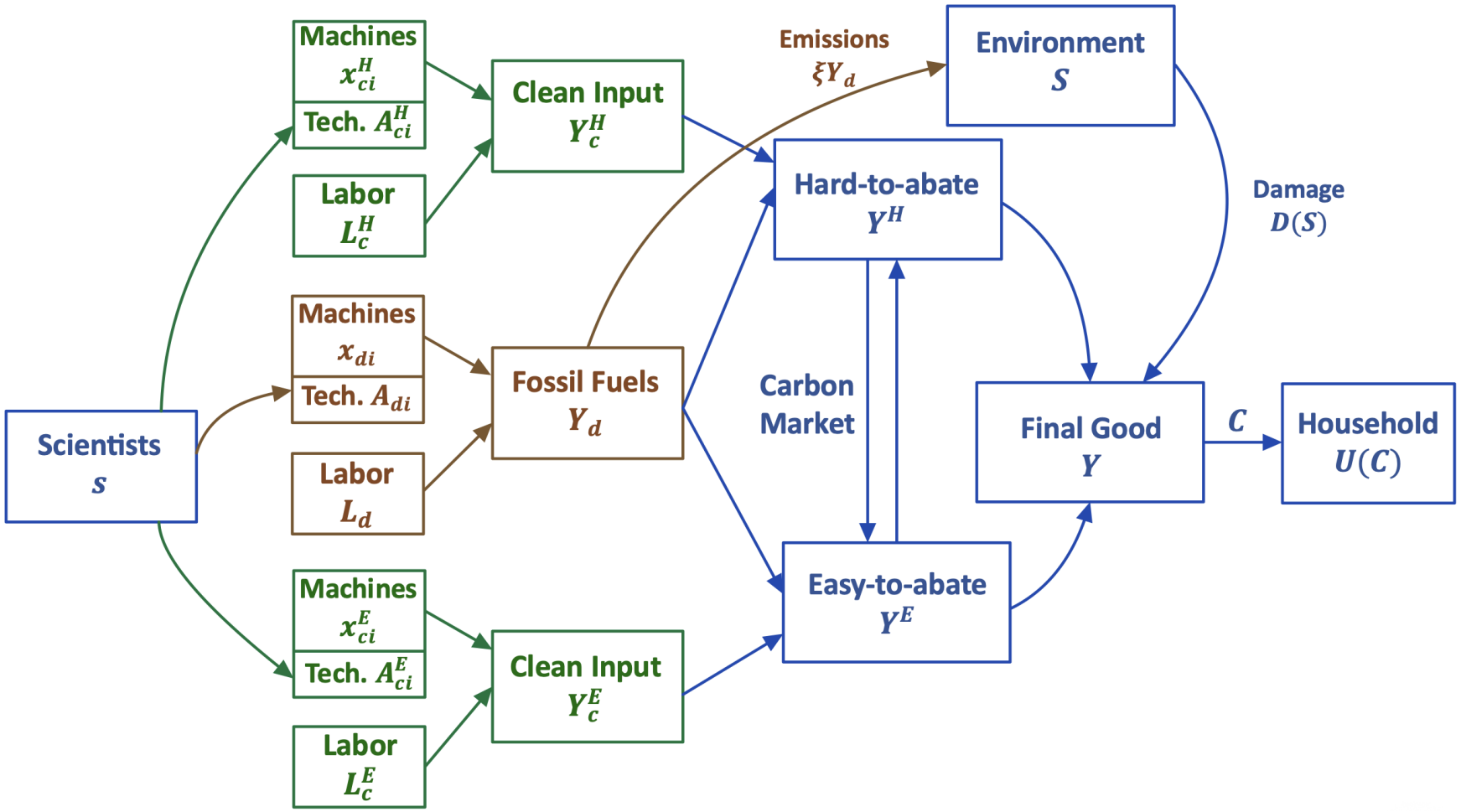


Note: This figure plots the estimated β_k coefficients from the following event study specification:

$$green_{it} = \sum_{k=-8}^{12} \beta_k \cdot I\{t = k\} \cdot ETS_i \cdot Hard_i + \sum_{k=-8}^{12} \gamma_k \cdot I\{t = k\} \cdot ETS_i + \delta \cdot x_{it} + \phi_i + \theta_{ct} + \lambda_{jt} + \varepsilon_{it}$$

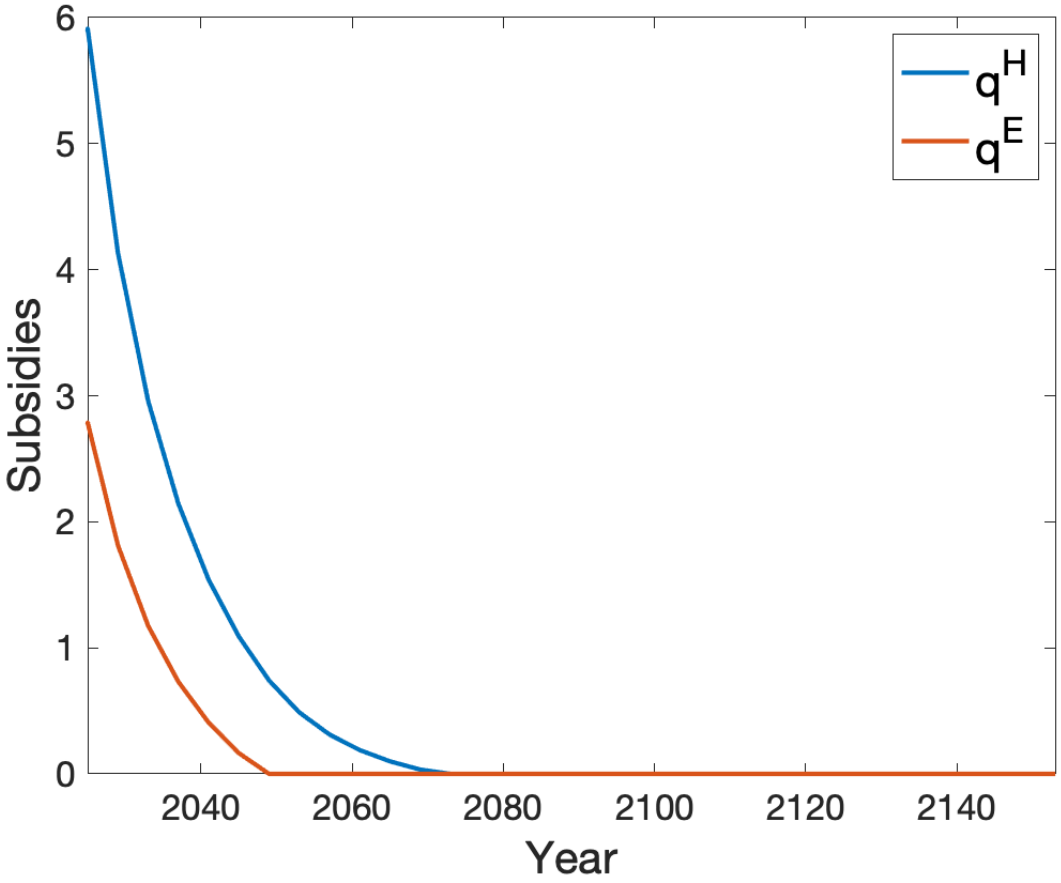
The outcome variable $green_{it}$ is an indicator equal to 1 if firm i files at least one green patent in year t. ETS_i is set to 1 for firms that are regulated by the EU-ETS (that is, operate at least one installation that participates in the EU-ETS), and $Hard_i$ is set to 1 if the firm operates in a hard-to-abate industry. $I\{t = k\} \cdot ETS_i \cdot Hard_i$ are year-relative-to-treatment indicator variables interacted with $ETS_i \cdot Hard_i$, and $I\{t = k\} \cdot ETS_i$ are year-relative-to-treatment indicator variables interacted with ETS_i. The omitted year is $k = -1$. The regression includes firm fixed effects (ϕ_i), country-year fixed effects (θ_{ct}), and industry-year fixed effects (λ_{jt}), and controls for the lagged free allocation ratio (x_{it}). Standard errors are clustered at the firm level. The treatment year ($k = 0$) is 2008 for all industries apart from chemicals, which joined the EU-ETS in 2013 (and $k = 0$ in year 2013).

Figure 4: Model Scheme



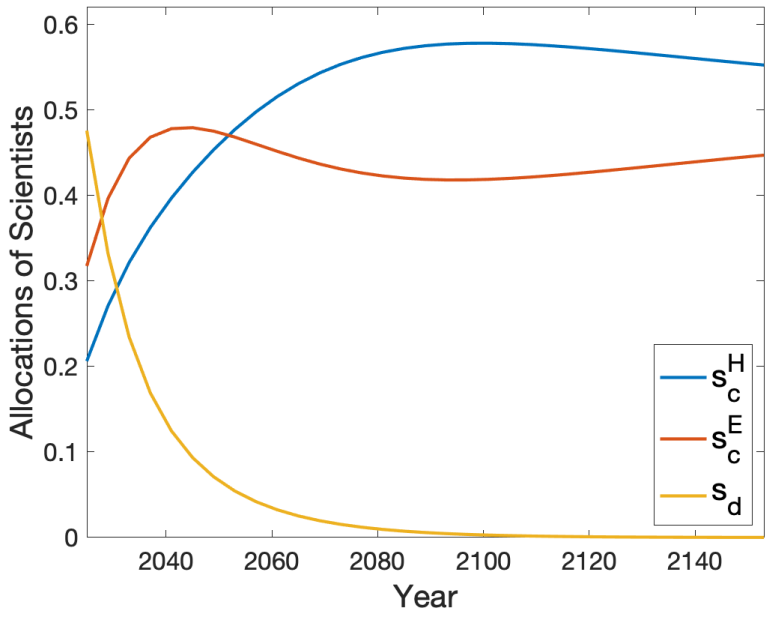
Note: This figure presents a scheme of the model. Complete details are in Section 5.

Figure 5: Optimal Subsidies

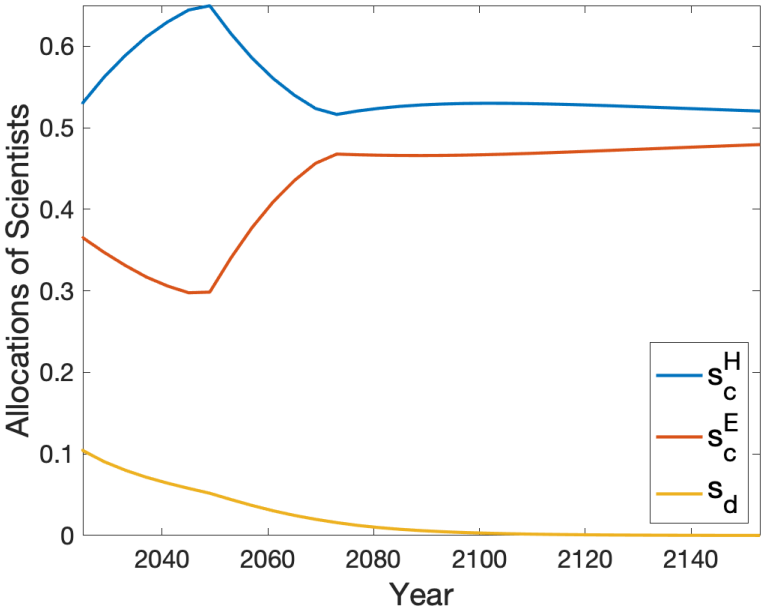


Note: This figure presents the path of optimal subsidies to scientists innovating in the clean hard-to-abate input sector (q^H) and the clean easy-to-abate input sector (q^E). The parameter values used are in Table 7.

Figure 6: Transition Dynamics - Scientists



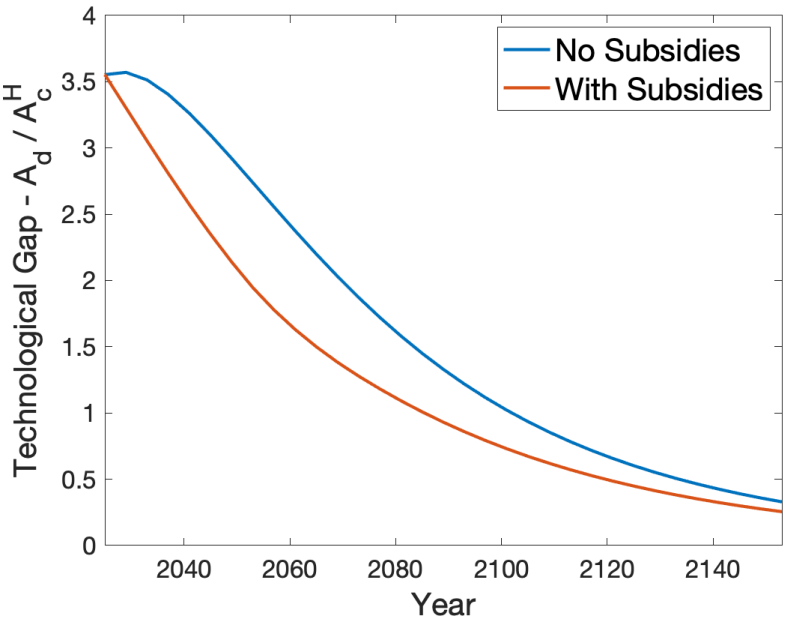
(a) No Subsidies



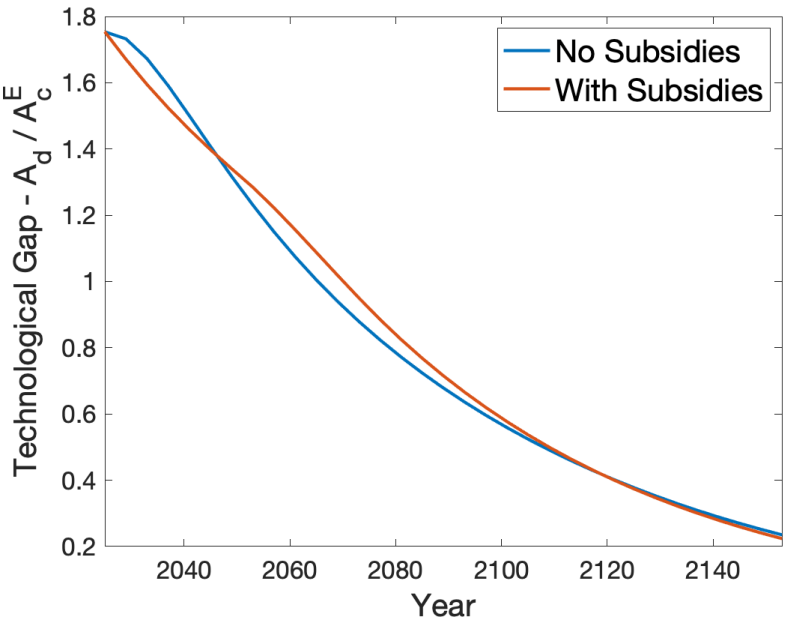
(b) Optimal Subsidies

Note: This figure presents the paths of the allocations of scientists innovating in the clean hard-to-abate input sector (s_c^H), the clean easy-to-abate input sector (s_c^E) and the dirty input sector (s_d). Panel A and Panel B present the paths under the no subsidies and optimal subsidies scenarios, respectively. The parameter values used are in Table 7.

Figure 7: Transition Dynamics - Technological Gaps



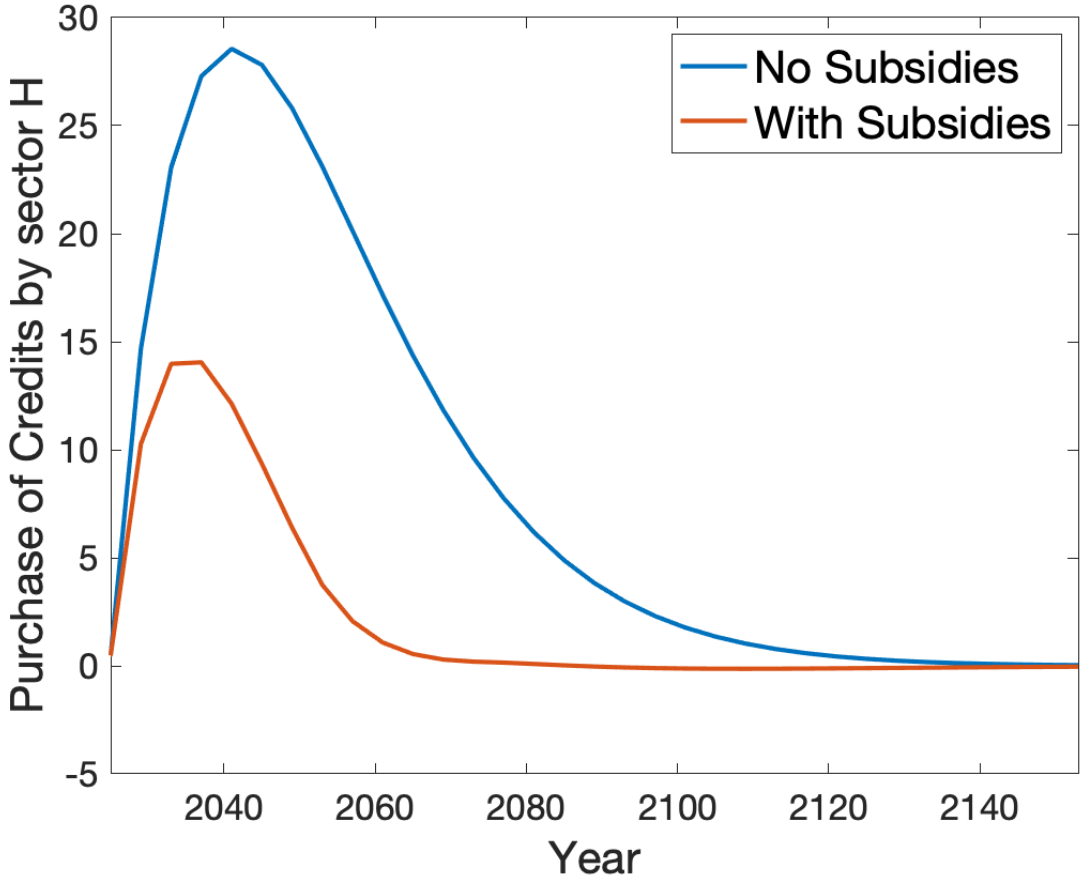
(a) No Subsidies



(b) Optimal Subsidies

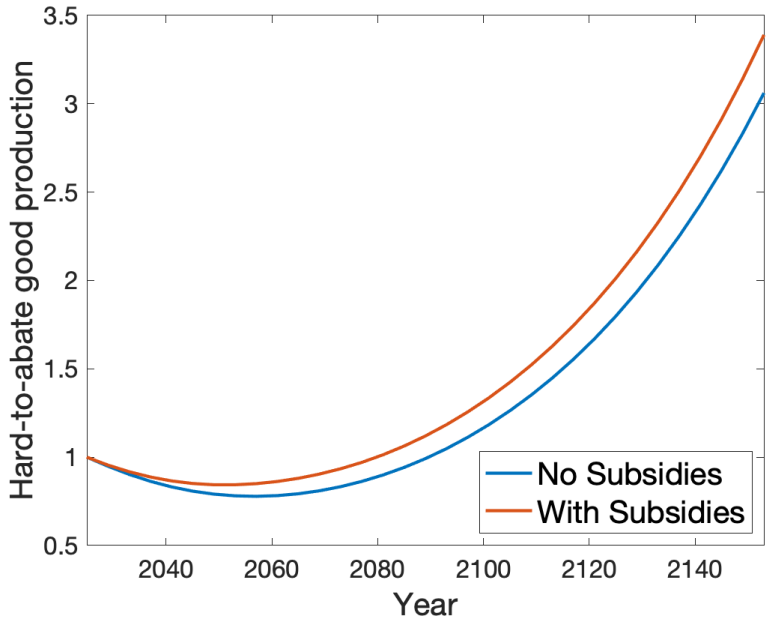
Note: This figure presents dynamics of the technological gaps between the dirty and sector specific clean technology. Panel A (Panel B) presents the dynamics of the gap between the dirty and the clean hard-to-abate (easy-to-abate) inputs under the no-subsidies and optimal subsidies cases. The parameter values used are in Table 7.

Figure 8: Emissions Trading

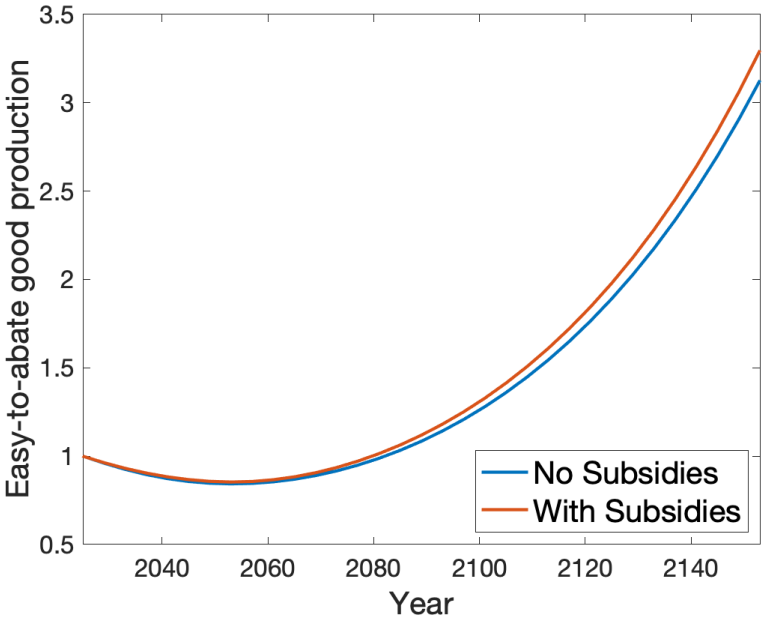


Note: This figure presents the path of emission allowances purchased by hard-to-abate firms under the no-subsidies and optimal-subsidies scenarios, given the parameter values in Table 7.

Figure 9: Production of Intermediate Goods



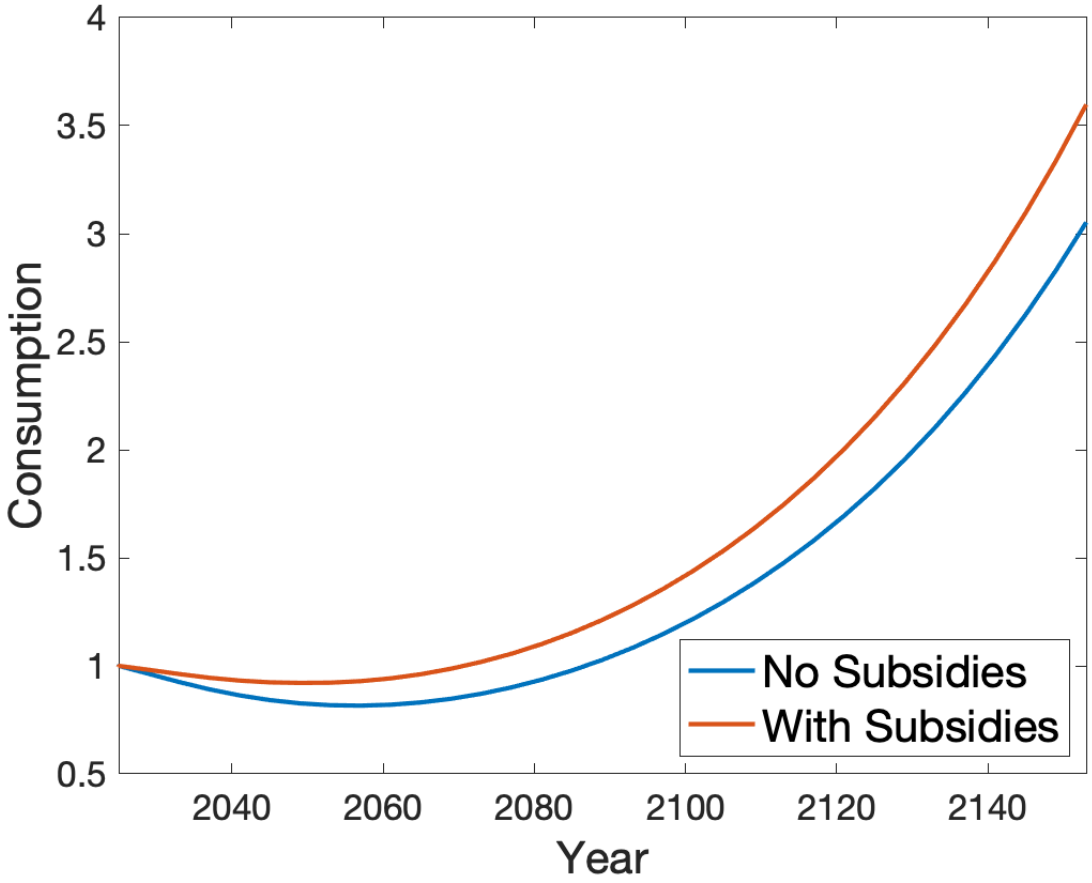
(a) Hard-to-abate (Y^H) Good Production



(b) Easy-to-Abate (Y^E) Good Production

Note: This figure presents the paths of the production of the hard-to-abate intermediate good Y^H (Panel A) and the easy-to-abate intermediate good Y^E (Panel B), normalized by the respective 2021 levels, comparing the no subsidies and optimal subsidies scenarios. The parameter values used are in Table 7.

Figure 10: Effect of Subsidies on Household Consumption



Note: This figure presents the dynamics of household consumption C, normalized by the 2021 level of consumption, comparing the no subsidies and optimal subsidies scenarios. The parameter values used are in Table 7.

A Extended Technological Background

While policy reports indicate that the prices of renewable energy keep falling (International Renewable Energy Agency, 2022), hard-to-abate industries face stubborn emissions reduction challenges. For example, a recent report by the US Department of Energy (Brooks, 2022) shows that the US has a renewable energy technical potential to produce more than 100 times its electricity consumption with renewable resources, and that "every region of the country has access to multiple renewable energy resources".[25] On the other hand, reducing emissions in the hard-to-abate industries requires substantial technological advancements, and 35% of emissions reductions in the IEA's Net Zero Emissions by 2050 Scenario come from technologies that are not yet developed (International Energy Agency, 2021). It is common to classify GHG emissions into two types: *heat emissions*, which are generated from the combustion of fossil fuels, and *process emissions*, in which GHGs are emitted due to chemical processes as part of production. Hard-to-abate industries face decarbonization challenges in both types. In the following section I discuss the core challenges the three largest hard-to-abate industries face.

A.1 Steel Industry

Steel production, which accounts for 7-9% of global CO_2 emissions (World Steel Association, 2021), involves two main steps. First, iron ore (Fe_2O_3) is smelted in a blast furnace at temperatures of around 900°C with a carbon-rich substance, mainly coke (C), in order to separate iron (Fe) from oxygen (O):

$$_2Fe_2O_3 + 3\,C \longrightarrow 4\,Fe + 3\,CO_2$$

Emissions are generated because oxygen binds with carbon to form CO_2, which is released as a by-product, and because this process requires extremely high temperatures, achieved by combustion of fossil fuels.[26] The product, pig iron, is then processed and alloyed with other elements to produce steel with the desired properties. Thus, decarbonizing the industry requires technological alternatives to pig iron smelting.[27] Two

[25] Renewable energy technical potential "[...] represents the achievable energy capacity and generation of a particular technology given the resource potential, system performance, topographic limitations, environmental constraints, and land use constraints." (Cox et al., 2018)

[26] https://www.energy.gov/eere/iedo/iron-and-steel-manufacturing (Accessed on 01/29/2025).

[27] https://www.iea.org/energy-system/industry/steel (Accessed on 01/29/2025).

such prominent early-stage methods are Low Temperature Electrolysis and Molten Oxide Electrolysis. Additionally, hydrogen-based reduction, often referred to as "green steel", smelts iron ore using hydrogen produced from renewable energy, with water (H_2O) as the sole by-product. These low-emission technologies still need to be perfected and scaled up in order for the steel industry to decarbonize.[28] As the World Steel Association states in a policy report: "...to achieve the drastic [emissions] reductions needed, an entirely new, transformative approach to ironmaking is required and there are several promising initiatives under development." (World Steel Association, 2021).

A.2 Cement Industry

The cement industry accounts for roughly 8% of global CO_2 emissions (Ellis et al., 2020). More than half of these are *process* emissions generated when limestone ($CaCO_3$) is decomposed in a kiln at around 1,450°C to produce calcium oxide (CaO), emitting CO_2:

$$CaCO_3 \longrightarrow CaO + CO_2$$

The rest of the emissions are *heat* emissions, resulting from combustion of fossil fuels to achieve such high temperatures.[29] The US Department of Energy notes that "to help tackle the challenge of decarbonizing the cement industry, research and development (R&D) is needed across the following areas: (1) Next-generation cement/concrete formulations and production routes; (2) Low-carbon fuels and (3) Carbon-capture technologies" (An example of (1) is production of cement from calcium silicate, which will eliminate *process* emissions[30]).[31]

A.3 The Chemical Industry

The chemical industry accounts for about 5% of global CO_2 emissions (Gabrielli et al., 2023). The main challenge in decarbonizing production of chemicals is that over half of the energy input is used as a feedstock, with fossil fuels supplying carbon and hydrogen as raw materials. Therefore, it requires inventing low-emitting substitutes for the emissions-intensive inputs. Accordingly, key milestones in the energy transition of the

[28] https://www.iea.org/energy-system/industry/steel (Accessed on 01/29/2025)
[29] https://www.energy.gov/eere/iedo/cement-and-concrete-manufacturing (Accessed on 01/29/2025).
[30] https://www.iea.org/energy-system/industry/cement (Accessed on 01/29/2025)
[31] https://www.energy.gov/eere/iedo/cement-and-concrete-manufacturing (Accessed on 01/29/2025).

chemicals industry include using biofuels and waste gases as inputs and increasing the recyclability of chemical products.[32]

B Additional Empirical Results and Robustness Checks

Minimum Pre-policy Patenting Requirement Appendix Table A.2 examines the robustness of the main result to different cutoffs of pre-policy patenting activity. The baseline restricts attention to firms that filed at least one EPO patent before 2008 (Column 1). Column 2 removes this requirement entirely, including all firms in the sample regardless of pre-policy patenting; the triple-interaction estimate weakens to -0.028 (significant at the 1% level), as expected when adding firms that are less engaged in innovation. Columns 3 to 5 instead tighten the requirement to firms with at least two, three, and five pre-policy EPO patents. The estimate strengthens monotonically, from -0.063 to -0.10, consistent with the differential being concentrated among firms with established innovative capacity. Across all sample definitions the triple-interaction estimate remains negative and significant at the 1% level.

Joint Estimation of the Sectoral Emissions Paths In the main text I estimate the emissions paths of easy- and hard-to-abate installations as two separate event studies (Figure 1). In this appendix I instead estimate the two sectors together, in a single regression. Doing so offers two advantages. First, it allows me to formally test whether the emissions reductions of the two sectors are statistically different at each horizon. Second, by including calendar-year fixed effects, it absorbs shocks common to both sectors, such as the 2008-2009 recession, so that the estimated gap reflects their differential evolution rather than aggregate fluctuations. I therefore pool the two sectors and interact event time with a hard-to-abate indicator:

$$\log(emissions_{it}) = \alpha_i + \gamma_t + \sum_{\tau=1}^{10} \beta_\tau \, I\{\text{event time}_{it} = \tau\} + \sum_{\tau=1}^{10} \delta_\tau \left(\text{Hard}_i \times I\{\text{event time}_{it} = \tau\}\right) + \varepsilon_{it}, \tag{44}$$

where α_i denotes installation fixed effects, γ_t calendar-year fixed effects, and Hard_i equals one for hard-to-abate installations. As before, event time counts the years since installation i's inclusion, with the inclusion year ($\tau = 0$) as the omitted baseline, and standard

[32] https://www.energy.gov/eere/iedo/chemical-and-petrochemical-manufacturing (Accessed on 01/29/2025).

errors are clustered at the installation level. The coefficients β_τ trace the easy-to-abate path, and δ_τ measures the difference between the two sectors at each event time.

Figure A.1 plots this difference, computed as $100\,(\exp(-\delta_\tau) - 1)$ with 95% confidence intervals. A negative value implies that easy-to-abate installations cut emissions by more. The difference is small and insignificant in the first two years, then turns negative and widens from the third year onward, ranging from about 17% to 33% between years three and nine and significant over most of this horizon. This confirms that the divergence shown in Figure 1 is statistically significant.

C Model Appendix

C.1 Are Carbon Markets Active in Equilibrium?

Whether intermediate good producers trade emission allowances in equilibrium depends on the stringency of the emissions caps $\bar{Y}^k_{dt}$. I analyze the three potential equilibrium cases, of which only two are feasible:

1. **None of the constraints bind - No trade:** For all $k \in \{E, H\}$, $Y^k_{dt} < \bar{Y}^k_{dt} + Z^k_t$.
   In this case, the first-order conditions (FOCs) of Equation 24 imply that $p_{zt} = 0$, resulting in no trade of emission allowances. Thus, $Z^H_t = Z^E_t = 0$.

2. **Both constraints bind - Active trade:** For all $k \in \{E, H\}$, $Y^k_{dt} = \bar{Y}^k_{dt} + Z^k_t$.
   Here, intermediate good producers actively trade emission allowances. By the FOCs of Equation 24, $p_{zt} > 0$, and the carbon market clearing condition $Z^H_t + Z^E_t = 0$ determines the price. One sector (as shown later, the easy-to-abate) is a net seller of allowances, while the other is a net buyer (the hard-to-abate).

3. **The constraint binds only for one sector (not feasible):** $Y^k_{dt} = \bar{Y}^k_{dt} + Z^k_t$ for some $k \in \{E, H\}$ while $Y^{-k}_{dt} < \bar{Y}^{-k}_{dt} + Z^{-k}_t$.
   This scenario cannot constitute an equilibrium. Suppose it can. Then, the FOCs of Equation 24 for sector $-k$ imply that $p_{zt} = 0$ since the constraint does not bind. However, with $p_{zt} = 0$ (the shadow price of the emissions constraint for sector k), the constraint of sector k also cannot bind, leading to a contradiction.

C.2 Equilibrium Solution

We start with the input producers. In each sector $k \in \{E, H\}$, clean input producers choose the number of machines x_{cit}^k and labor L_{ct}^k to maximize profits:

$$\max_{x_{cit}^k, L_{ct}^k} \left\{ p_{ct}^k (L_{ct}^k)^{1-\alpha} \int_0^1 (A_{cit}^k)^{1-\alpha} (x_{cit}^k)^\alpha di - w_t L_{ct}^k - \int_0^1 p_{cit}^k x_{cit}^k di \right\} \tag{45}$$

which yields the following demand function: $x_{cit}^k = \left(\frac{\alpha p_{ct}^k}{p_{cit}^k} \right)^{\frac{1}{1-\alpha}} A_{cit}^k L_{ct}^k$.
Analogously, dirty input producers choose the number of machines x_{dit} and labor L_{dt} to maximize profits:

$$\max_{x_{dit}, L_{dt}} \left\{ p_{dt} (L_{dt})^{1-\alpha} \int_0^1 (A_{dit})^{1-\alpha} (x_{dit})^\alpha di - w_t L_{dt} - \int_0^1 p_{dit} x_{dit} di \right\} \tag{46}$$

which yields the following demand function: $x_{dit} = \left(\frac{\alpha p_{dt}}{p_{dit}} \right)^{\frac{1}{1-\alpha}} A_{dit} L_{dt}$.

The first-order conditions with respect to labor of each of the three problems yield the following relations between relative levels of technology and relative prices:

$$\frac{p_{ct}^k}{p_{dt}} = \left(\frac{A_{ct}^k}{A_{dt}} \right)^{-(1-\alpha)}, \quad \text{for each } k \in \{E, H\} \tag{47}$$

The monopolistic producer of clean machine i in intermediate good sector $k \in \{E, H\}$ chooses p_{cit}^k and x_{cit}^k to maximize profits subject to the demand function above:

$$\max_{x_{cit}^k, p_{cit}^k} \pi_{cit}^k = \left(p_{cit}^k - \psi \right) x_{cit}^k \quad s.t. \quad x_{cit}^k = \left(\frac{\alpha p_{ct}^k}{p_{cit}^k} \right)^{\frac{1}{1-\alpha}} A_{cit}^k L_{ct}^k \tag{48}$$

Using the normalization that $\psi = \alpha^2$, we derive that $p_{cit}^k = \alpha$. Thus, equilibrium profits are

$$\pi_{cit}^k = (1-\alpha)\alpha A_{cit}^k L_{ct}^k (p_{ct}^k)^{\frac{1}{1-\alpha}} \tag{49}$$

and the demand for machines can be written as:

$$x_{cit}^k = A_{cit}^k L_{ct}^k (p_{ct}^k)^{\frac{1}{1-\alpha}} \tag{50}$$

Combining 8, 16 and 50, we obtain the equilibrium supply of clean input c in intermediate good sector k:

$$Y_{ct}^k = A_{ct}^k L_{ct}^k \left(p_{ct}^k\right)^{\frac{\alpha}{1-\alpha}} \tag{51}$$

Analogously, we obtain that dirty input supply is:

$$Y_{dt} = A_{dt} L_{dt} p_{dt}^{\frac{\alpha}{1-\alpha}} \tag{52}$$

As to the demand side, intermediate good producers in each sector $k \in \{E, H\}$ choose input quantities, Y_{ct}^k and Y_{dt}^k, and amount of emission allowances purchased, Z_t^k, to maximize profits subject to the policy constraint:

$$\max_{Y_{ct}^k, Y_{dt}^k, Z_t^k} \left\{ p_t^k Y_t^k - p_{ct}^k Y_{ct}^k - p_{dt} Y_{dt}^k - p_{zt} Z_t^k \right\} \quad s.t. \quad Y_{dt}^k \leq \bar{Y}_{dt}^k + Z_t^k \tag{53}$$

This problem yields the following ratio between input demands and input and carbon prices, implying that relative demand of the cheaper input is higher:

$$\frac{Y_{ct}^k}{Y_{dt}^k} = \left(\frac{p_{ct}^k}{\tilde{p}_{dt}}\right)^{-\varepsilon} \tag{54}$$

where $\tilde{p}_{dt}$ is the *effective* price of the dirty input and is defined as $\tilde{p}_{dt} \equiv p_{dt} + p_{zt}$. Importantly, as the carbon price p_{zt} increases, relative demand of the dirty input decreases. We also obtain the following relation between *effective* input prices and the price of intermediate good k:

$$\left((\tilde{p}_d^k)^{1-\varepsilon} + (p_c^k)^{1-\varepsilon}\right)^{\frac{1}{1-\varepsilon}} = p^k \tag{55}$$

and that $Y_{dt} \leq \bar{Y}_{dt} + Z_t^k$. I discuss below the two equilibrium cases: (1) no-trade and (2) active trade cases.

Final good producers choose intermediate good quantities, Y_t^H and Y_t^E, to maximize profits:

$$\max_{Y_t^E, Y_t^H} \left\{ Y_t - p_t^E Y_t^E - p_t^H Y_t^H \right\} \tag{56}$$

which yields the following relation between intermediate goods demands and prices ratios:

$$\frac{Y_t^E}{Y_t^H} = \left(\frac{p_t^E}{p_t^H}\right)^{-\nu} \tag{57}$$

and the following intermediate good price relation:

$$(p^H)^{1-\nu} + (p^E)^{1-\nu} = 1 \tag{58}$$

Note that the price of the final good is normalized to 1.

C.3 Scientists' Equilibrium Decision

This section proves that in the unique equilibrium, the allocation of scientists $s^* \equiv (s_{dt}^*, s_{ct}^{H*}, s_{ct}^{E*})$ is such that scientists innovate in all sectors $s^* \in (0,1)^3$ and expected profits are equal across sectors. I start by showing that there exists a unique allocation $s^* \in (0,1)^3$, with $s_{ct}^{H*} + s_{ct}^{E*} + s_{dt}^* = 1$ for all $t = 1, .., \infty$ such that Equation 31 holds. I then show that this is an equilibrium. The proof is concluded by showing that this equilibrium is unique.

Existence of $s^* \in (0,1)^3$, with $s_{ct}^{H*} + s_{ct}^{E*} + s_{dt}^* = 1$ for all $t = 1, .., \infty$:

Define $X_{ct}^k \equiv \eta_c \alpha(1-\alpha) p_{ct}^k Y_{ct}^k, k = H, E$ and $X_{dt} \equiv \eta_d \alpha(1-\alpha) p_{dt} Y_{dt}$. By Equation 31, define x such that:

$$x \equiv (1 + \frac{\gamma}{s_{ct}^H}) X_{ct}^H = (1 + \frac{\gamma}{s_{ct}^E}) X_{ct}^E = (1 + \frac{\gamma}{s_{dt}}) X_{dt} \tag{59}$$

Combining this with scientists' market clearing condition, we obtain the following equation with one unknown, x:

$$\frac{\gamma X_{ct}^H}{x - X_{ct}^H} + \frac{\gamma X_{ct}^E}{x - X_{ct}^E} + \frac{\gamma X_{dt}}{x - X_{dt}} = 1 \tag{60}$$

Define $f(x)$ to be:

$$f(x) = \frac{\gamma X_{ct}^H}{x - X_{ct}^H} + \frac{\gamma X_{ct}^E}{x - X_{ct}^E} + \frac{\gamma X_{dt}}{x - X_{dt}} - 1 \tag{61}$$

By $s^* \in [0,1]^3$ (which follows from (1) the share of scientists in each sector is non-negative; and (2) market clearing), we have

$$x \in X_{ct}^H \cdot (1+\gamma, \infty) \cap X_{ct}^E \cdot (1+\gamma, \infty) \cap X_{dt} \cdot (1+\gamma, \infty) \tag{62}$$

implying that $x \geq max\{X_{ct}^H(1+\gamma), X_{ct}^E(1+\gamma), X_{dt}(1+\gamma)\} \equiv \underline{x}$.
Showing that there exists $x^* \in (\underline{x}, \infty)$ such that $f(x^*) = 0$ will conclude this part of the proof, because the definition of x above implies a unique allocation of scientists:

$$s_{ct}^E = \frac{\gamma X_{ct}^E}{x - X_{ct}^E}, \quad s_{ct}^H = \frac{\gamma X_{ct}^H}{x - X_{ct}^H}, \quad s_{dt} = \frac{\gamma X_{dt}}{x - X_{dt}} \tag{63}$$

To that end, I show that (1) $f(x)$ is continuous and strictly decreasing for $x \in (\underline{x}, \infty)$, (2) $f(\underline{x}) > 0$, and (3) $\lim_{x\to\infty} f(x) < 0$, and use the Intermediate Value Theorem to conclude that there exists $x^* \in (\underline{x}, \infty)$ such that $f(x^*) = 0$.

1. It is clear to see that $f(x)$ is continuous for $x \in (\underline{x}, \infty)$. It is also strictly decreasing:

$$f'(x) = -\frac{\gamma X_{ct}^H}{(x - X_{ct}^H)^2} - \frac{\gamma X_{ct}^E}{(x - X_{ct}^E)^2} - \frac{\gamma X_{dt}}{(x - X_{dt})^2} < 0 \tag{64}$$

2. Suppose without loss of generality that $\underline{x} = X_{dt}(1+\gamma)$. Thus:

$$f(\underline{x}) = \frac{\gamma X_{ct}^H}{X_{dt}(1+\gamma) - X_{ct}^H} + \frac{\gamma X_{ct}^E}{X_{dt}(1+\gamma) - X_{ct}^E} + \frac{\gamma X_{dt}}{X_{dt}(1+\gamma) - X_{dt}} - 1 = \tag{65}$$

$$= \frac{\gamma X_{ct}^H}{X_{dt}(1+\gamma) - X_{ct}^H} + \frac{\gamma X_{ct}^E}{X_{dt}(1+\gamma) - X_{ct}^E} + 1 - 1 > 0 \tag{66}$$

since $X_{dt} > X_{ct}^H$ and $X_{dt} > X_{ct}^E$.

3. When $x \to \infty$:

$$lim_{x\to\infty} f(x) = lim_{x\to\infty} \left(\frac{\gamma X_{ct}^H}{x - X_{ct}^H} + \frac{\gamma X_{ct}^E}{x - X_{ct}^E} + \frac{\gamma X_{dt}}{x - X_{dt}} \right) - 1 = -1 \tag{67}$$

By the Intermediate Value Theorem, there exists a unique $x^* \in (\underline{x}, \infty)$ such that $f(x^*) = 0$. Therefore, there exists $s^* \in (0,1)^3$ such that $s_{ct}^{H*} + s_{ct}^{E*} + s_{dt}^* = 1$ for all $t = 1, .., \infty$ and Equation 31 holds.

Why is s^* an equilibrium?

The scientist allocations that satisfy Equation 31 equalize expected pay in all sectors. Given that s^* satisfies Equation 31, no scientist has an incentive to deviate and innovate in another sector since it would yield a lower expected pay, because $\Gamma(\cdot)$ is a decreasing function and $s_{ct}^{H*} + s_{ct}^{E*} + s_{dt}^* = 1$. Therefore, these allocations constitute an equilibrium.

Uniqueness of the equilibrium

Suppose there is another allocation of scientists $s' \equiv (s_{ct}^{H'}, s_{ct}^{E'}, s_{dt}') \in (0,1)^3$, with $s_{ct}^{H'} + s_{ct}^{E'} + s_{dt}' = 1$ that is an equilibrium. By uniqueness of s^* that satisfy Equation 31, it must be that expected pay in at least one of the sectors is higher. However, this cannot be an equilibrium since scientists in the sector(s) with lower expected pay will switch to the sector(s) with higher. Finally, we need to show that there is no equilibrium for which $s = 0$ for at least some $s \in \{s_{ct}^H, s_{ct}^E, s_{dt}\}$ (which also implies that there is no $s = 1$ for one $s \in \{s_{ct}^H, s_{ct}^E, s_{dt}\}$), which will allow us to conclude that the unique equilibrium is the one we found above. There is no such $s = 0$ for some $s \in \{s_{ct}^H, s_{ct}^E, s_{dt}\}$ because expected pay in the corresponding sector will be ∞ (since $\lim_{s \to 0+} \Gamma(s) = \infty$), and scientists innovating in other sectors will receive higher expected pay from switching to this sector.

Appendix Tables

Table A.1: CPC Y02 Technology Classification

CPC Code	Technology Description	Category
Y02B	**Climate change mitigation tech. related to buildings**	
Y02B 10/00	Integration of renewable energy sources in buildings	Easy
Y02B 20/00	Energy efficient lighting tech.	Easy
Y02B 30/00	Energy efficient heating, ventilation or air conditioning [HVAC]	Easy
Y02B 90/00	Enabling tech. or tech. with a potential or indirect contribution to GHG emissions mitigation	Easy
Y02C	**Capture, storage, sequestration or disposal of GHG**	Hard
Y02D	**Climate change mitigation tech. in information and communication tech. [ICT]**	Easy
Y02E	**Reduction of GHG emissions, related to energy generation, transmission or distribution**	-
Y02E 10/00	Energy generation through renewable energy sources	Easy
Y02E 20/00	Combustion tech. with mitigation potential	Easy
Y02E 30/00	Energy generation of nuclear origin	Easy
Y02E 40/00	Tech. for an efficient electrical power generation, transmission or distribution	Easy
Y02E 50/00	Tech. for the production of fuel of non-fossil origin	Hard
Y02E 60/00	Enabling tech.; Tech. with a potential or indirect contribution to GHG emissions mitigation	-
Y02E 60/10	Energy storage using batteries	Easy
Y02E 60/13	Energy storage using capacitors	Easy
Y02E 60/14	Thermal energy storage	Easy
Y02E 60/16	Mechanical energy storage, e.g. flywheels or pressurised fluids	Easy
Y02E 60/60	Arrangements for transfer of electric power between AC networks or generators via a high voltage DC link [HVDC]	Easy
Y02E 70/00	Other energy conversion or management systems reducing GHG emissions	Easy
Y02P	**Climate change mitigation tech. in the production or processing of goods**	-
Y02P 10/00	Tech. related to metal processing	Hard
Y02P 20/00	Tech. relating to chemical industry	Hard
Y02P 30/00	Tech. relating to oil refining and petrochemical industry	Hard
Y02P 40/00	Tech. relating to the processing of minerals	Hard
Y02P 60/00	Tech. relating to agriculture, livestock or agroalimentary industries	
Y02P 60/80	Food processing, e.g. use of renewable energies or variable speed drives in handling, conveying or stacking	Easy
Y02P 90/40	Fuel cell tech. in production processes	Hard
Y02P 90/45	Hydrogen tech. in production processes	Hard
Y02P 90/60	Electric or hybrid propulsion means for production processes	Easy
Y02P 90/70	Combining sequestration of CO2 and exploitation of hydrocarbons by injecting CO2 or carbonated water in oil wells	Hard

Note: This table presents a classification of CPC Y02 patent categories into technologies that are predominantly related to easy-to-abate or hard-to-abate economic activities. Codes whose Category is marked "-" are classified at a more granular level.

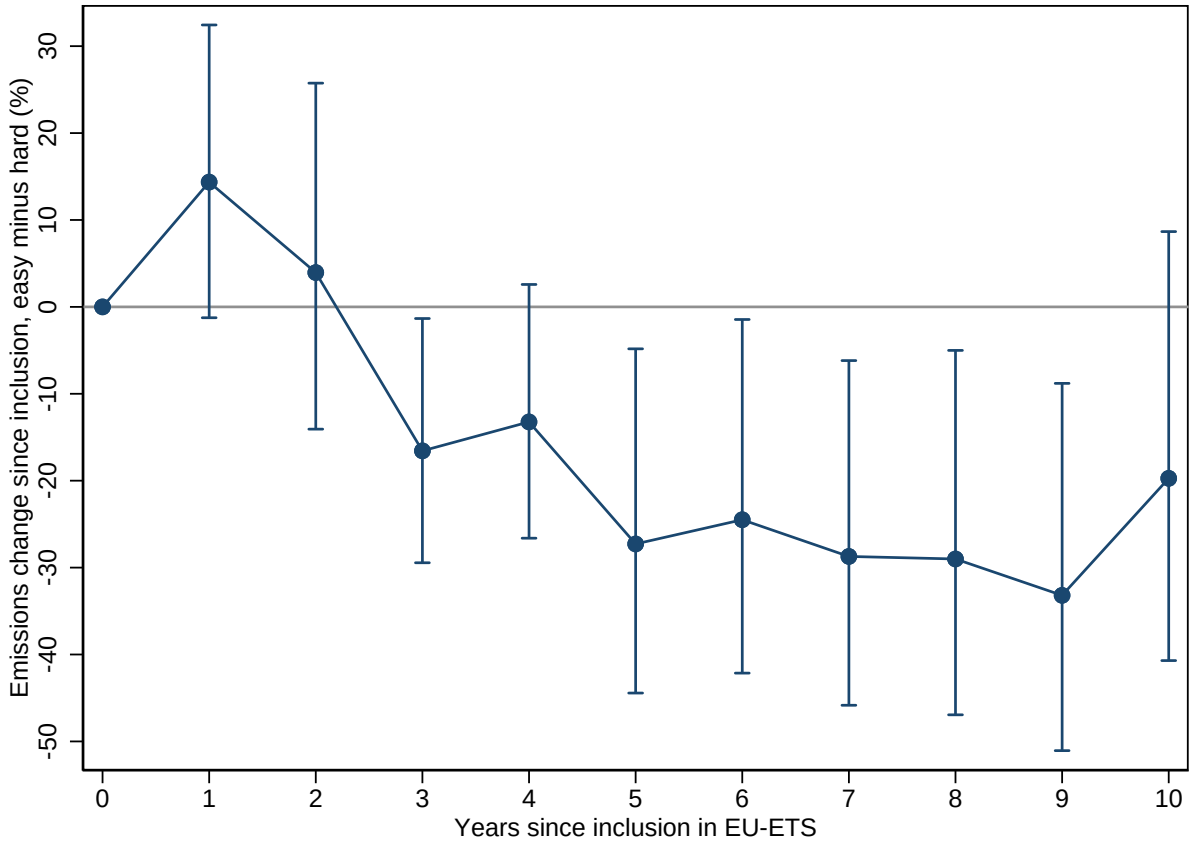
Table A.2: Robustness Checks - Minimum Pre-Policy Patenting Requirement

	(1) Green patenting	(2) Green patenting	(3) Green patenting	(4) Green patenting	(5) Green patenting
Post × ETS	0.069***	0.033***	0.081***	0.099***	0.12***
	(0.018)	(0.0091)	(0.023)	(0.026)	(0.034)
Post × ETS × Hard	-0.063***	-0.028***	-0.070***	-0.085***	-0.10***
	(0.020)	(0.0100)	(0.026)	(0.029)	(0.037)
Sample	Baseline (≥1 pre-policy patent)	No requirement (≥0)	≥2 pre-policy patents	≥3 pre-policy patents	≥5 pre-policy patents
Firm FE	Y	Y	Y	Y	Y
Country × Year FE	Y	Y	Y	Y	Y
Industry × Year FE	Y	Y	Y	Y	Y
Uncond. Mean	0.028	0.012	0.039	0.047	0.060
Observations	36,981	100,039	24,886	19,794	14,638
Adj. R-squared	0.43	0.39	0.44	0.45	0.47

Note: This table presents triple-difference regression results on engagement in green innovation under alternative pre-policy patenting requirements on the estimation sample. The dependent variable is an indicator variable set to 1 if the firm filed an own-technology green patent in that year, defined as a green patent in a technology relevant to the firm's own easy- or hard-to-abate activity (Section 3.2). The "Sample" row reports the pre-policy patenting requirement: Column (1) is the baseline sample (firms that filed at least one EPO patent before 2008, the pre-policy period); Column (2) imposes no pre-policy patenting requirement; and Columns (3)-(5) restrict the sample to firms that filed at least two, three, and five pre-policy patents, respectively. All columns include firm, country-year and industry-year fixed effects and control for the lagged emissions-to-free-allocation ratio and the lagged log patent stock. Standard errors (in parentheses) are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

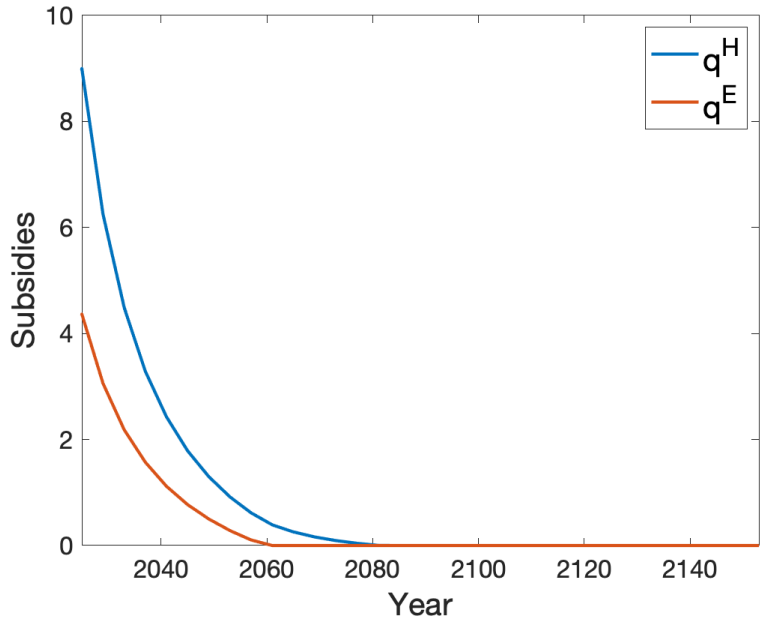
Appendix Figures

Figure A.1: Difference in emissions change: easy- minus hard-to-abate installations

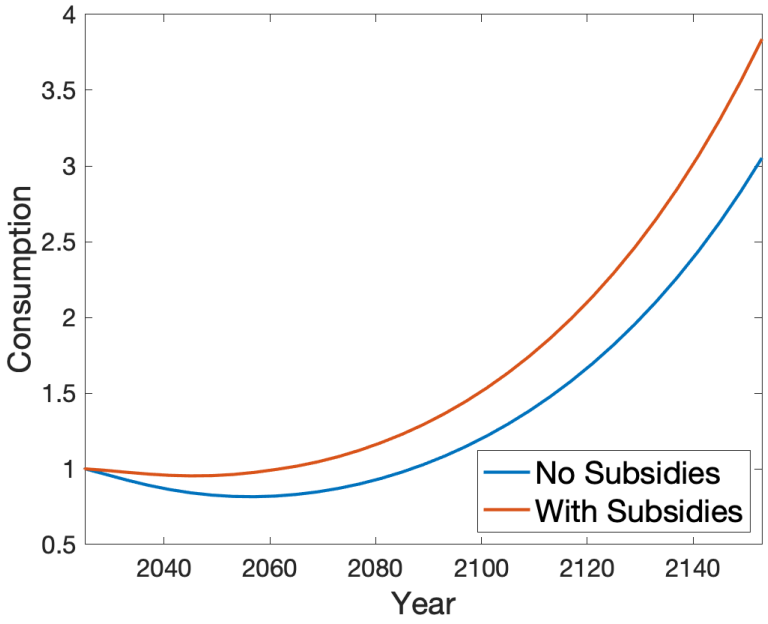


Note: This figure plots the estimated easy-minus-hard emissions differential from the joint event study in Equation (44). The plotted series is $-\delta_\tau$, the difference in log emissions between easy- and hard-to-abate EU-ETS installations at each event time relative to the year of inclusion ($\tau = 0$), expressed as an implied percentage change, $100 \cdot (\exp(-\delta_\tau) - 1)$, with 95% confidence intervals. A negative value indicates that easy-to-abate installations reduced emissions by more. The regression includes installation fixed effects (α_i) and calendar-year fixed effects (γ_t). The event-time main effect at $\tau = 10$ is omitted owing to the collinearity between event time, calendar year, and year of inclusion. The sample includes all installations operated by main sample firms, as defined in Section 3.2, that participate in the EU-ETS. Standard errors are clustered at the installation level.

Figure A.2: Robustness Check - Low Pure Rate of Time Preference $\rho = 0.1\%$



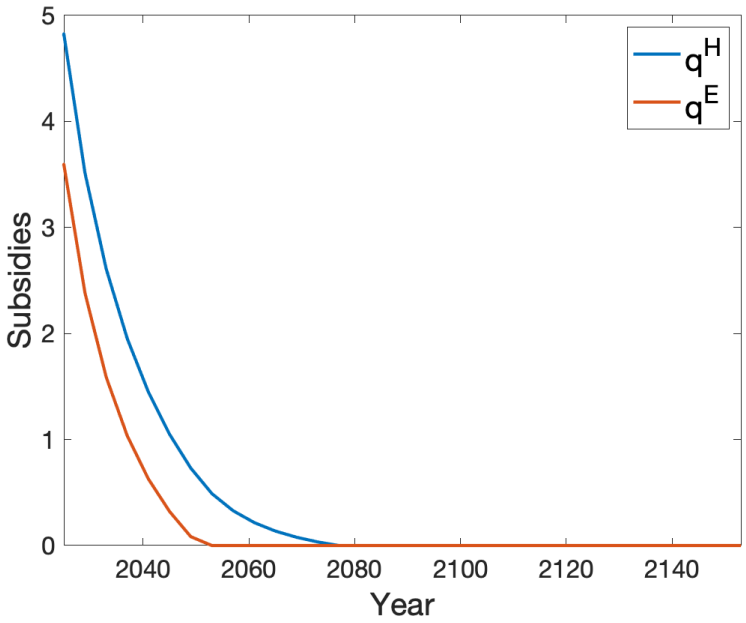
(a) Optimal Subsidies



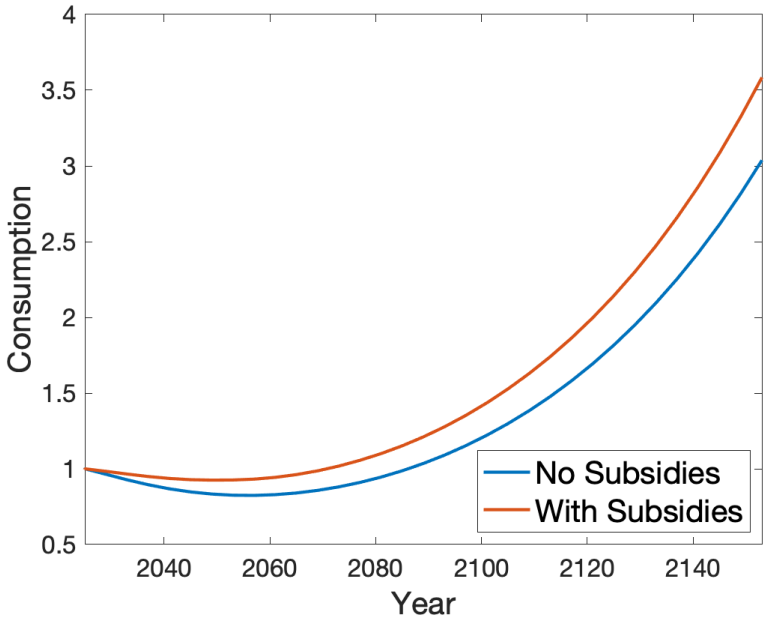
(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a lower value of the pure rate of time preference $\rho = 0.1\%$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.

Figure A.3: Robustness Check - High Intermediate Goods Substitution $\nu = 0.75$



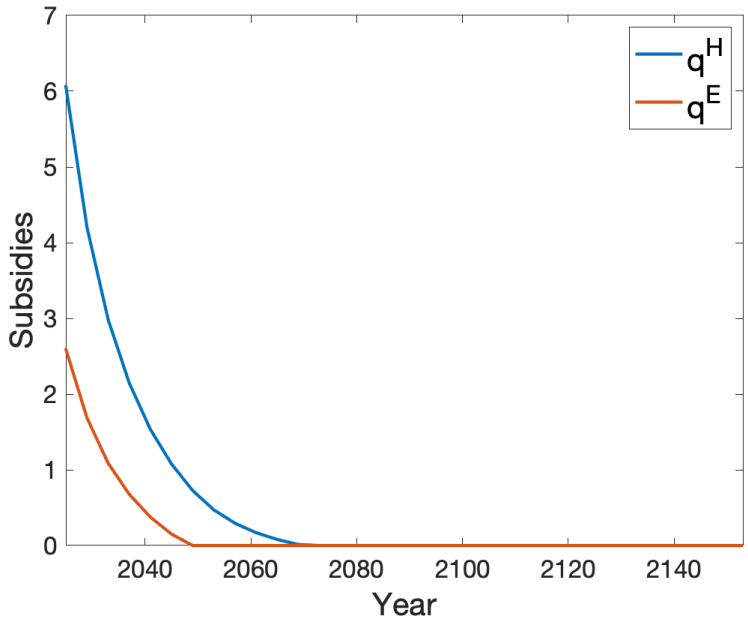
(a) Optimal Subsidies



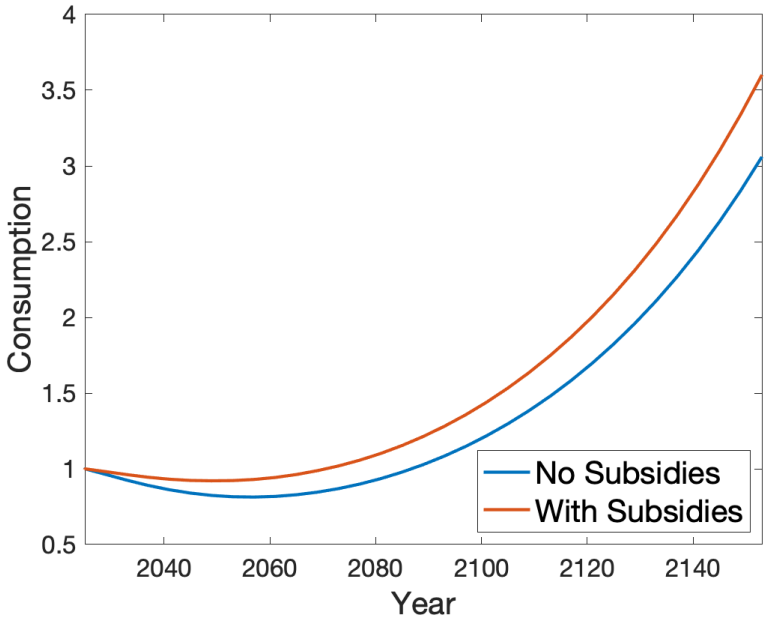
(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a higher value of the elasticity of substitution of intermediate goods $\nu = 0.75$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.

Figure A.4: Robustness Check - Low Intermediate Goods Substitution $\nu = 0.2$



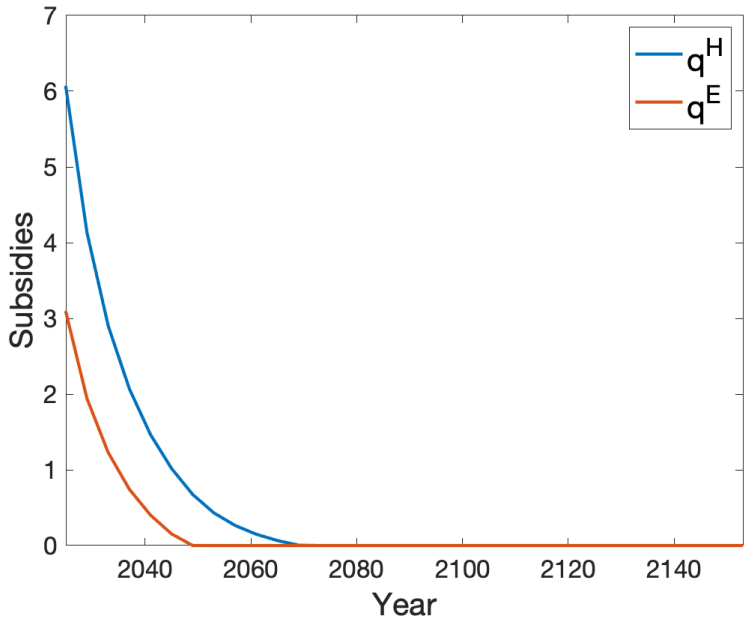
(a) Optimal Subsidies



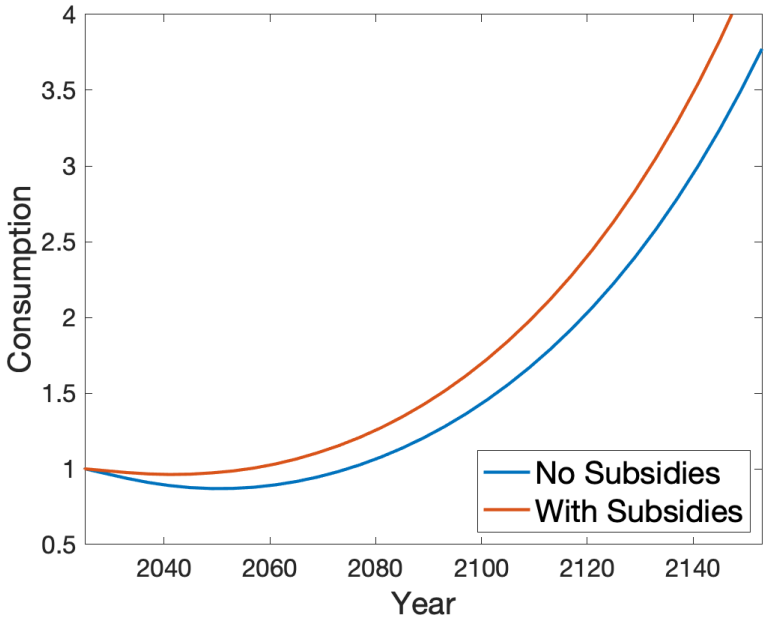
(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a lower value of the elasticity of substitution of intermediate goods $\nu = 0.2$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.

Figure A.5: Robustness Check - High Input Elasticity of Substitution $\varepsilon = 2.87$



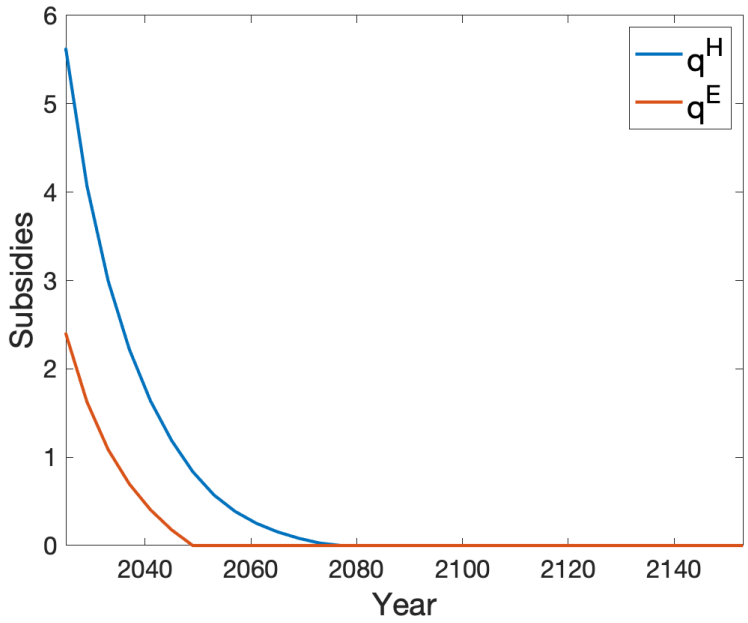
(a) Optimal Subsidies



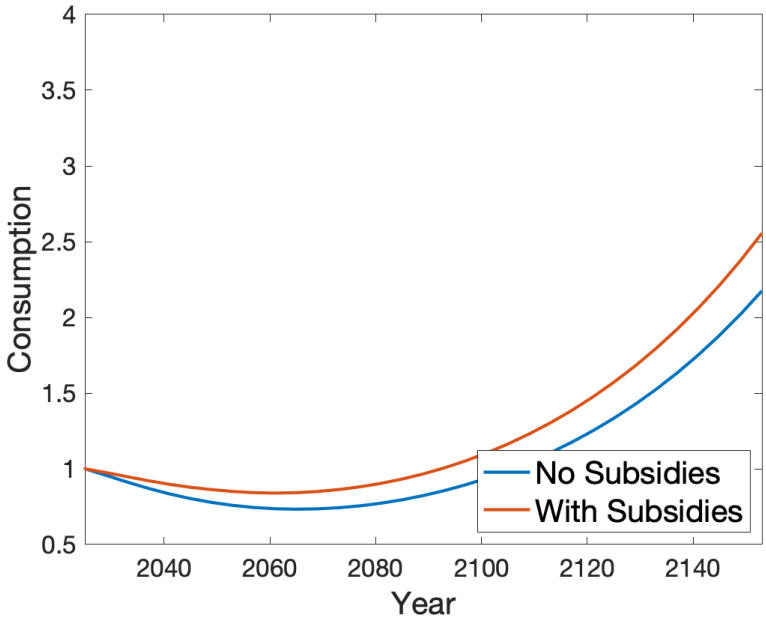
(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a higher value of the elasticity of substitution between the inputs $\varepsilon = 2.87$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.

Figure A.6: Robustness Check - Low Input Elasticity of Substitution $\varepsilon = 2.05$



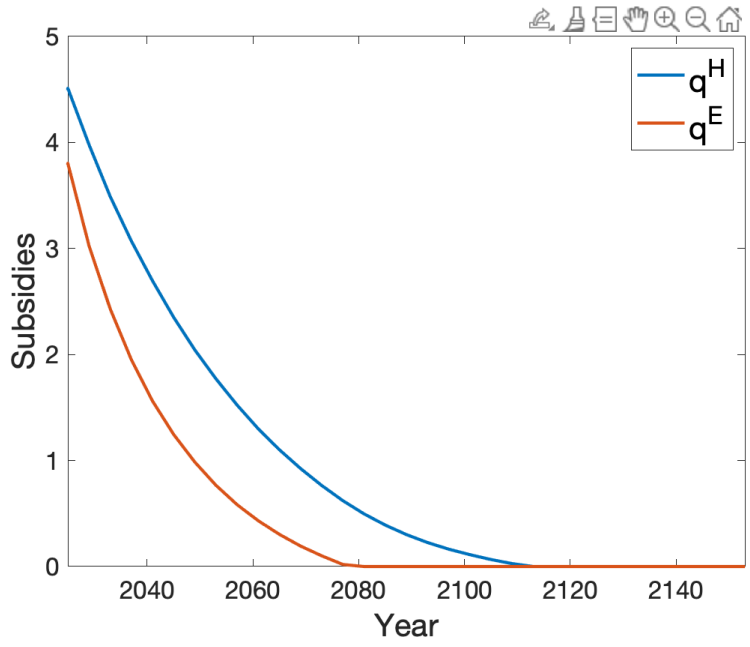
(a) Optimal Subsidies



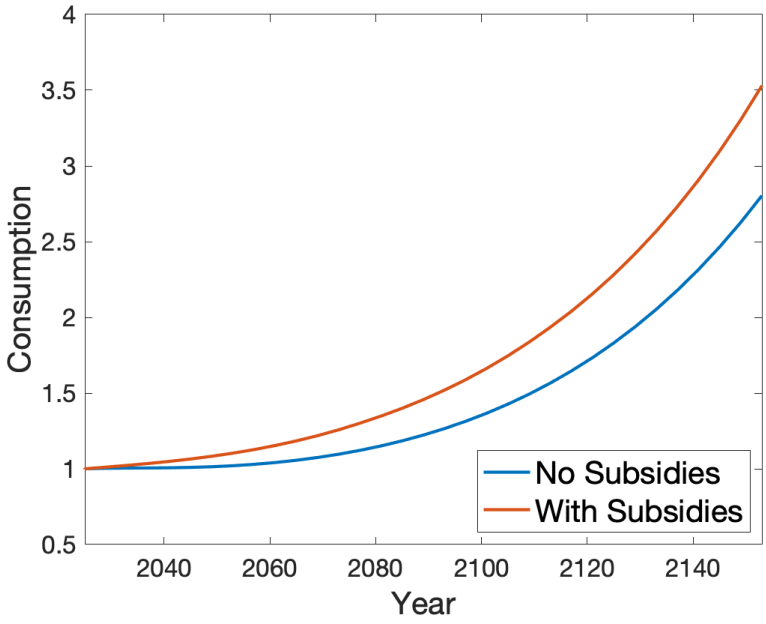
(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a lower value of the elasticity of substitution between the inputs $\varepsilon = 2.05$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.

Figure A.7: Robustness Check - Low Emissions Cap Decrease Rate $\theta = 2.2\%$



(a) Optimal Subsidies



(b) Effect of Subsidies on Household Consumption

Note: This figure presents the main results of the calibration, conducting a robustness check to a lower value of the emissions decrease rate $\theta = 2.2\%$. The rest of the parameter values are in Table 7. Panel A presents the optimal path of subsidies to scientists innovating in the clean hard-to-abate (easy-to-abate) sector q^H (q^E), and Panel B presents the effect of subsidies on household consumption.